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Structural Vector Auto
Regression (SVAR)
Comparison of the United
States and United Kingdom

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Honours Dissertation
Declaration



This dissertation is submitted in partial fulfilment of the requirements for the degree of BA with Honours in the Strathclyde Business School.

I declare that this dissertation embodies the results of my own work and that it has been composed by myself. Following normal academic conventions, I have made due acknowledgement to the work of others.

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Abstract

This dissertation examines whether the Bank of England's inflation-targeting framework anchored bond market expectations more effectively than the Federal Reserve's dual mandate during the post-2020 inflation shock. Using country-specific Structural Vector Auto Regressions identified through sign restrictions, this study compares how supply-driven inflation shocks transmit to 10-year inflation expectations and government bond yields in the United States and United Kingdom for the period March 1996 to December 2025. The full-sample results confirm that both central banks maintained credible inflation anchors, with supply-shocks explaining less than 7% of long-term expectation variance in either country. However, a regime-shift analysis, splitting the sample at January 2020, reveals that the UK's pre-COVID anchoring advantage eroded substantially. Before 2020, supply shocks drove 7.71% of UK expectation variance compared to 11.91% in the US. After 2020, UK sensitivity rose to 12.73% while the US figure remained essentially unchanged at 12.60%. The UK's institutional advantage disappeared under the compound stress of a global supply shock, energy dependence, and post-Brexit supply frictions. At the same time, UK gilt yields exhibited structurally higher sensitivity to supply shocks than US Treasuries across the full sample time period, pointing to an elevated inflation risk premium in UK sovereign debt. The findings suggest that while explicit inflation targeting anchors expectations effectively under normal conditions, this insulation is not unconditional when sufficiently large real-world shocks interact with country-specific structural vulnerabilities.

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Chapter 1: Introduction

The relationship between inflation and government bond yields is widely studied in macroeconomics, yet the post-2020 inflation shock has exposed weaknesses in the effectiveness of that research when central bank credibility is actively being tested. Between 2021 and 2022, consumer price inflation in both the United States (US) and the United Kingdom (UK) reached levels not seen in four decades (Goryunov et al., 2023). Both countries were affected by similar causes: pandemic supply disruptions, energy price shocks following Russia's invasion of Ukraine, and the delayed effects of extraordinary monetary stimulus. Both the Federal Reserve and the Bank of England responded with historically aggressive tightening cycles. The question this dissertation investigates is whether those responses were perceived to be equally credible by the bond markets.

This question is important because bond yields do not simply reflect expected inflation. They also include a risk premium, the extra compensation investors demand for the possibility that inflation turns out higher than expected, which erodes the real value of their holdings. When a central bank is credible, this premium stays low because markets trust the bank will bring inflation back to target effectively. When credibility decreases, the premium rises, pushing up borrowing costs for governments, firms, and households regardless of what the central bank does next. The impact of lost credibility is not abstract. It is reflected in gilt prices and Treasury yields within hours of a data release.

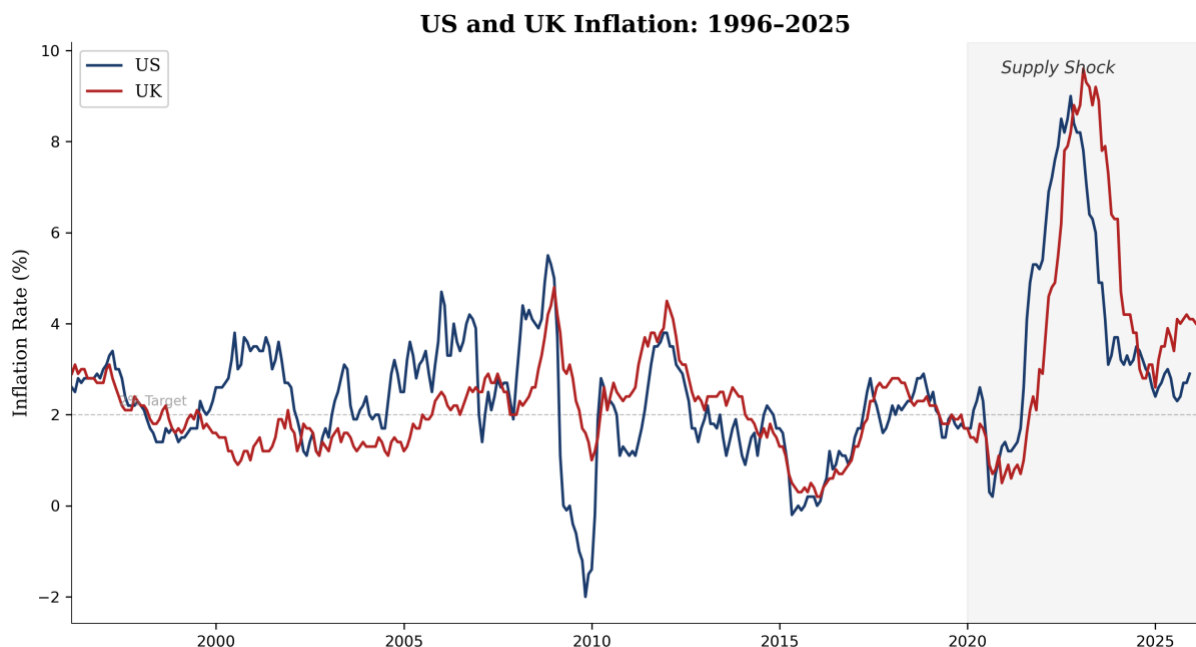
A substantial amount of pre-2020 research documented a transatlantic divergence in how effectively each country anchored inflation expectations. Gürkaynak et al. (2005) showed that the Bank of England's explicit inflation-targeting framework, adopted in 1992, insulated UK long-term forward rates from macroeconomic news more effectively than the Federal Reserve's dual mandate, which historically tolerated greater ambiguity about the long-run price level. This finding has become embedded in the macro-finance literature, but the evidence was drawn almost entirely from the Great Moderation, a period of unusual macroeconomic stability that ended with the 2007 Global Financial Crisis and has arguably not returned since.

The post-2020 environment offers a good comparison to benchmark against. Both countries faced a common global supply shock of similar magnitude. Yet their institutional responses, structural vulnerabilities, and bond market outcomes differed in ways that the pre-existing literature does not fully explain. For example, the UK's greater reliance on imported energy

magnified domestic price pressures. Post-Brexit trade frictions also introduced supply-side pressures with no US equivalent and the Bank of England’s tightening cycle, although aggressive, was widely perceived as more hesitant than the Federal Reserve’s front-loaded approach (Bailey, 2024).

This dissertation addresses whether the Bank of England’s pre-2020 anchoring advantage survived the post-2020 stress test, or whether the “Global Financial Cycle” (Miranda-Agrippino and Rey, 2020) overwhelmed domestic institutional frameworks entirely. Country-specific Structural Vector Autoregressions (SVARs) are undertaken using US and UK monthly data from March 1996 to December 2025 to address the question. The models are identified using sign restrictions, following Uhlig (2005) which distinguish supply-driven (“bad”) inflation shocks from demand-driven (“good”) ones. This specific distinction matters because bond markets respond differently to each (Cieślak and Pflueger, 2023). Impulse response functions (IRFs) and Forecast Error Variance Decompositions (FEVD) are compared across a pre-COVID baseline (1996–2019) and a post-COVID stress-test regime (2020–2025), providing a direct before-and-after comparison of central bank credibility.

Figure 1: US and UK Year-on-Year Inflation, March 1996 – December 2025. Source: FRED (US CPI); ONS (UK CPIH). Grey shading highlights the post-2020 supply shock period.



This study contributes to the existing literature in several ways. It provides one of the first direct comparative tests of the anchoring hypothesis using post-2020 data, a period that most

of the foundational studies in this field do not cover (see literature review section). The sign restriction identification avoids the “Price Puzzle” that compromises recursive Cholesky approaches (Sims, 1992), allowing a cleaner separation of supply and demand shocks than standard VAR methods permit. Finally, the regime-shift analysis explicitly tests whether the structural break identified by Reis (2022), who argued that the 2021–2022 inflation surge represents a genuine regime change rather than a transitory spike, has altered the credibility of the both the US and UK central banks as priced in sovereign bond markets.

The findings point in a clear direction. Over the full sample, both anchors hold: supply shocks explain less than 7% of variance in long-term expectations for both countries. But the regime-shift analysis reveals a significant narrowing. Pre-2020, the Bank of England held a measurable credibility advantage over the Federal Reserve, consistent with Gürkaynak et al. (2005). However, after 2020, this advantage largely disappeared. UK expectation sensitivity to supply shocks increased by two-thirds, while the US figure remained essentially unchanged. The anchor did not fully break, but the direction of the shift is unambiguous.

Chapter 2 reviews the literature and identifies the gaps this study addresses. Chapter 3 describes the SVAR framework and data. Chapter 4 presents the results. Chapter 5 concludes, discussing limitations and directions for further research.

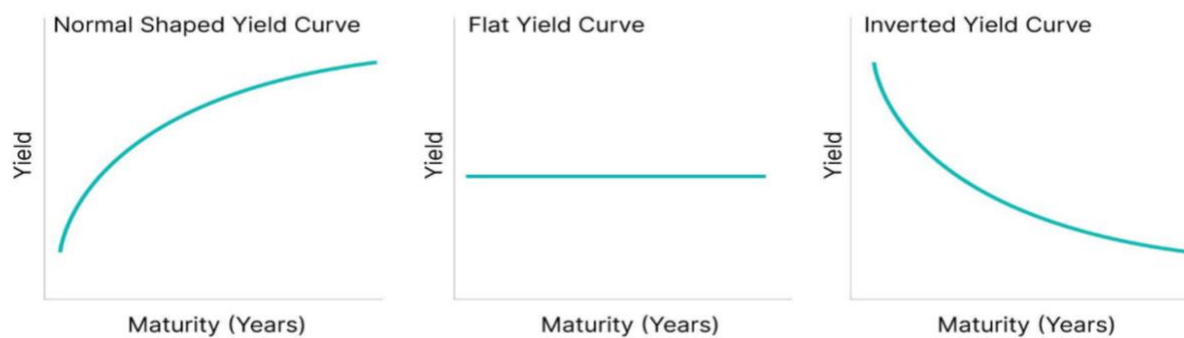
Chapter 2: Literature Review

2.1 Theoretical Foundations: The Macro-Finance Evolution

The interaction between inflation shocks and government bond markets remains one of the most scrutinized relationships in macroeconomics. In this chapter, the theoretical and empirical literature underpinning the relationship between inflation and government bond yields is reviewed. Starting with an examination the Macro-Finance consensus and the development of vector autoregression (VAR) methodologies for the term structure. The US and UK experiences are then contrasted, focusing on the role of central bank credibility in anchoring expectations. Finally, a gap in the literature is identified namely: the need for a comparative VAR analysis that accounts for the structural breaks introduced by the recent resurgence of global inflation following the 2020 COVID pandemic disruption of the markets.

The term structure of interest rates, commonly visualised as the yield curve, describes the relationship between bond yields and their time to maturity (Figure 2). The classical Fisher hypothesis serves as the theoretical foundation and suggests nominal bond yields should adjust one-for-one with predicted inflation to preserve real returns (Fisher, 1930). However, the reality is much more complex. News about expected inflation typically accounts for only 10% to 20% of shocks to nominal Treasury yields, leaving the remainder to be explained by the durability of shocks, the legitimacy of central bank aims, and the fluctuating risk premiums that investors require to keep government debt (Duffee, 2018). This is important in addressing the aims of this dissertation because it means a significant share of the yield movement observed after an inflation shock reflects risk premiums and credibility perceptions, and this is precisely the channels compared between the US and UK.

Figure 2: Stylised yield curve shapes. The normal upward-sloping curve (left) reflects expectations of future growth, the flat curve (middle) signals economic uncertainty, and the inverted curve (right) has historically preceded recessions.



Historically, the term structure was modelled using latent factor models (Level, Slope, and Curvature), which explain nearly all yield variation (Litterman and Scheinkman, 1991). These three factors correspond to the distinct shape a yield curve can take in practice (Figure 2). These models are not considered to be based on theory, instead they are descriptive showing how yields move but not explaining why (Diebold et al., 2006). The Macro-Finance revolution, led by Ang and Piazzesi (2003), bridged this gap by introducing No-Arbitrage Vector Autoregression (NA-VAR), these models explicitly combine economic theory and statistical modelling. No-arbitrage models assume that investors cannot earn a risk-free profit by trading between bonds of different maturities, which forces yields at each maturity to be consistent with the same underlying expectations. This constraint allows the model to link macroeconomic shocks directly to the shape of the yield curve, rather than treating yields at different maturities as independent. Their findings were transformative: they showed that while unobserved factors still dominate the "long end" of the yield curve, macroeconomic factors (inflation) explain up to 85% of the variation in short-term and medium-term yields.

At its core, the economic decomposition of nominal yields can be viewed as the sum of expected short-term real rates, expected inflation, and a term premium that compensates for both inflation and liquidity risks. Structural identification is required to move beyond the observed yield and isolate which of these components: the fundamental expectation or the risk premium, is driving the response to a macro shock. This decomposition provides the essential link between economic theory and the subsequent choice of an SVAR framework.

No-arbitrage VAR models reveal a common factor structure between core inflation and interest rates that consistently outperforms traditional survey forecasts (Ajello et al., 2020). However, these models are not without limitations: NA-VAR frameworks impose strict parameter stability assumptions that may not hold across different inflation regimes, a concern that is particularly acute when applying Great Moderation-era estimates to the post-2020 environment.

2.2 Measurement, Risk, and the Nature of Shocks

Understanding how an inflation shock propagates through the yield curve is particularly critical in the context of the Great Moderation (GM). GM was a period of unprecedented macroeconomic stability in advanced economies, spanning from approximately 1985 until the onset of the 2007 Global Financial Crisis. Characterized by a significant reduction in the volatility of both real GDP growth and inflation, this GM era provided the stable empirical environment in which foundational macro-finance models were developed. The parameters in these models were estimated during a period of unusual calm, so applying them to the post-COVID inflationary surge, which this dissertation does, risks systematic bias if the underlying shock dynamics have changed. This dissertation argues that the assumptions of parameter stability born from the GM period may no longer hold in the volatile, post-2020 COVID era. As Stock and Watson (2007) demonstrated, the stochastic properties of inflation have changed significantly over recent decades. They argued that inflation became harder to forecast using traditional backward-looking Phillips Curve models as monetary policy became more effective. This shift presents a challenge for researchers: if inflation follows a "random walk" or contains a large unobserved component, how do bond markets look forward and understand what drives yields?

To address this, contemporary research has divided into two distinct, but complementary, approaches. The first is the market-microstructure approach (e.g. Barr and Campbell, 1997) which utilises the unique structure of the UK government bond market, specifically the coexistence of nominal and index-linked gilts. Barr and Campbell (1997) demonstrated that inflation expectations could be extracted directly from asset prices, offering a high-frequency alternative to survey data. Their work underscored that bond yields contain not just expected inflation, but also varying risk premia that fluctuate in response to economic news.

The second approach, and the one central to this dissertation, is the macro-finance Vector Autoregression (VAR) framework. While the previous section established how these models work, it is worth noting their key limitation here: VAR models require the researcher to impose assumptions about ordering and persistence of shocks. Incorrect assumptions can produce misleading results (Sims, 1992). This means the choice of identification strategy is a substantive methodological decision in this thesis.

Comparing the United States and the United Kingdom through this lens offers an informative natural experiment. Despite their relative global stability, the sensitivity of long-term rates to economic news varies significantly between the two nations, largely due to their differing institutional histories regarding inflation targeting (Gürkaynak et al., 2005). While the UK adopted explicit inflation targeting early, helping to anchor expectations, the US Federal Reserve operated under a dual mandate that historically allowed for greater ambiguity.

Crucially, not all inflation shocks are equal in their bond market effects. Inflation can be described as "good" (demand-driven) or "bad" (supply-driven) (Cieślak and Pflueger, 2023), as a result asset prices and inflation risk premia respond fundamentally differently depending on the shock's origin. Supply shocks create a negative correlation between stock and bond returns, stripping Treasuries of their 'safe-haven' status and forcing a spike in the inflation risk premium. Furthermore market-based expectation measures can be distorted by liquidity premia, particularly during crises (Pflueger and Viceira, 2011).

A pure yield-curve model is structurally blind to shock origin (Evans and Marshall, 2007), suggesting that the SVAR approach is well positioned to separate these effects in a way that reduced-form factor models may not fully capture.

2.3 The Transatlantic Divergence

While the mechanism connecting inflation to bond yields is theoretically universal, evidence suggests a distinct structural divergence between the United States and the United Kingdom. This divergence is primarily accounted for due to the differing institutional histories of the Federal Reserve (US) and the Bank of England (UK), specifically regarding the adoption of inflation targeting.

Gürkaynak et al. (2005) provide a highly influential comparative study between the US and UK. Using high-frequency event studies, they measure how US and UK forward rates respond to macroeconomic news announcements. The contrast is striking. US, long-term forward rates moved significantly in response to surprise inflation and monetary policy announcements, implying that market participants routinely revised their long-run inflation views based on short-run data. In the UK, long-term rates were largely unresponsive to the same types of news. Gürkaynak et al. (2005) attributed this difference to the Bank of England's explicit inflation-targeting mandate, adopted in 1992, which gave markets a credible anchor that the Federal Reserve's dual mandate did not.

Under explicit inflation targeting, the central bank publicly commit to a specific inflation rate (e.g. 2% for the Bank of England). This gives bond markets a credible long-run anchor, investors know that even if prices spike today, the central bank will act to bring inflation back to target. This institutional difference is further illuminated by the unique market microstructure of the UK. As Barr and Campbell (1997) argued, the UK government's early issuance of index-linked gilts (since 1981) allowed for a more transparent transmission of information. In the UK, investors could hedge inflation risk directly, meaning that nominal yield shocks could be swiftly split into real rate shocks and inflation compensation. The US Treasury, by contrast, did not introduce Treasury Inflation-Protected Securities (TIPS) until 1997, almost 2 decades later, and the market remained illiquid for years, a structural illiquidity that likely amplified the excess sensitivity documented by Gürkaynak et al. (2005) in US data over the same period.

More recent research challenges the view that domestic frameworks isolate bond markets from external shocks. The Global Financial Cycle hypothesis, proposed by Miranda-Agrippino and Rey (2020), argues that US monetary policy serves as a global driver of risk aversion, affecting asset prices worldwide regardless of domestic exchange rate regimes. In simple terms, when the US Federal Reserve tightens policy, investors worldwide become more cautious and sell risky assets, including government bonds in smaller economies, regardless of what those countries' own central banks are doing. This aligns with Ehrmann and Fratzscher (2005), who found significant spillover effects where US bond market shocks drove movements in European yields, but the reverse was rarely true.

The post-2020 inflation episode provides an unusually clean test of the dominant drivers. While Gürkaynak et al. (2005) suggests that the UK's institutional framework insulates its yield curve

from inflation volatility, Miranda-Agrippino and Rey (2020) argue that the UK remains vulnerable to global (in particular US-driven) shocks. By employing a comparative VAR framework on post-2020 data, this study can test which force is dominant in the current high-inflation environment.

2.4 Identified Gaps

2.4.1 The Temporal Gap

The existing literature, despite its sophistication, has a significant gap: the applicability of established macro-finance models to the post-2020 economic landscape. The foundational studies reviewed in this chapter such as Stock and Watson (2007), Ang and Piazzesi (2003), and Gürkaynak et al. (2005) were largely predicated on data from the "Great Moderation" (~1985–2007). During this period, inflation was characterized by low volatility and high persistence, and central bank credibility was rarely tested by supply-side shocks of the magnitude seen recently.

As Reis (2022) argues, the inflation surge of 2021–2022 represents a structural break that challenges the "random walk" hypothesis of inflation. The return of supply-chain constraints and energy price shocks mirrors the volatility of the 1970s more closely than the 2000s, yet the financial architecture (inflation targeting, central bank independence) is vastly different. While recent studies have examined the post-2020 surge, there remains limited comparative high-frequency evidence exploring the US-UK relationship. If the regime has genuinely shifted, estimates derived from the Great Moderation data suggest a potential systematic bias, this study addresses this by focusing on regime instability within a transatlantic SVAR framework, a dimension that has received insufficient attention in broader panel-data analyses. Most recent studies focus on single-country analyses (e.g., Tenreyro, 2022 for the UK) or broad panel data, neglecting a direct, high-frequency comparison of how the US and UK bond markets have processed this return of volatility. Extending the sample to December 2025 addresses the gap identified by Reis (2022), who argued that the post-COVID inflation surge represents a regime change rather than a transitory spike. This allows for a formal test of whether the anchoring benefits of inflation targeting are permanent or conditional on the low-volatility environment of the Great Moderation.

Bernanke and Blanchard (2024), extending their earlier US-focused analysis to 11 economies, find that pandemic-era price shocks had more transient effects than comparable shocks in the

1970s, attributing this to more anchored inflation expectations and higher monetary policy credibility. Their cross-country evidence provides a benchmark against which the UK-specific erosion documented in this dissertation can be assessed.

2.4.2 The Methodological Gap

The identified methodological challenge concerns structural identification within the SVAR. Recursive (Cholesky) identification schemes impose a specific causal ordering on variables, yet this ordering is not without complexity: as Sims (1992) demonstrated, naive recursive SVARs generate the ‘Price Puzzle’ – the counterintuitive finding that a contractionary monetary shock raises inflation in the short run, arising from the model’s failure to capture forward looking information in the policy rate. Sign restriction approaches (Uhlig, 2005) sidestep this problem but introduce their own ambiguity in shock labelling. While sign restrictions effectively sidestep the ‘Price Puzzle’, it is important to acknowledge that they generate set-identification rather than point-identification meaning the results may depend on the specific design of the restrictions and can result in a range of admissible models. This dissertation acknowledges this limitation and will employ a Median Target Response (MTR) approach to select the most representative model from the identified set, ensuring the results are robust and not merely artifacts of the restriction design.

2.5 Next steps

This literature review synthesises three strands. The Macro-Finance Consensus validates VAR methodologies for yield curve analysis. The quantitative literature highlights inflation’s changing stochastic properties, necessitating models that allow for shifting volatility (Stock and Watson, 2007). The transatlantic divergence debate suggests institutional differences should lead to differing bond market sensitivities, though this insulation may be eroding in an increasingly globalised financial cycle.

Together, these three themes point to a single unresolved tension: whether the institutional framework of inflation targeting which Gürkaynak et al. (2005) credited with insulating UK bond markets from inflation volatility remains sufficient in a world where Miranda-Agrippino and Rey (2020) have shown that US monetary policy drives a global financial cycle that crosses institutional borders. This dissertation addresses this tension directly by applying a comparative SVAR to post-2020 data for the US and UK. It tests whether domestic anchoring or global

contagion dominated bond market dynamics during the largest inflation shock in four decades – the 2020 COVID pandemic. The following chapter outlines the identification strategy used to isolate these competing forces.

Chapter 3: Methodology and Data

3.1 Econometric Framework: Structural Vector Autoregression

In this dissertation a Structural Vector Autoregression (SVAR) framework is employed to evaluate how inflation shocks transmit to US and UK government bond yields. Latent factor models of the yield curve (Litterman and Scheinkman, 1991) can describe the statistical patterns in yield movements, but they cannot identify what drives them (Diebold et al., 2006). The SVAR addresses this limitation by following Ang and Piazzesi (2003) and integrating macroeconomic variables directly into the system, making it possible to isolate the effect of an inflation shock on the term structure rather than simply documenting co-movement.

If we consider a p -order reduced-form VAR for a four-variable system y_t :

$$y_t = c + A_1 y_{t-1} + \dots + A_p y_{t-p} + u_t$$

Where c is a vector of constants, $A_1, \dots, A_p$ are coefficient matrices, and u_t are reduced-form residuals with covariance matrix Σ . These residuals are correlated across equations, which means they cannot be given a structural interpretation directly, for instance a shock to inflation cannot be separated from a simultaneous shock to bond yields. To isolate individual structural shocks ε_t , we need a matrix B that maps the observed residuals into uncorrelated structural disturbances via $u_t = B\varepsilon_t$. The matrix B is constructed by taking the Cholesky decomposition P of the covariance matrix, Σ and rotating it on a randomly drawn orthogonal matrix Q , giving $B = PQ$. Each rotation produces a different candidate structural model. In the model only those candidates that satisfy the economic sign restrictions set out in Section 3.3 are retained.

An alternative to the SVAR is the Local Projection (LP) method of Jordà (2005), which estimates impulse response functions directly without imposing a VAR structure. LP estimators handle model misspecification better at long horizons, but they are less efficient in small samples and produce wider confidence intervals (Plagborg-Møller and Wolf, 2021). With only 62 observations in the post-2020 subsample, this loss of precision would make it difficult to distinguish between US and UK responses to supply shocks. The SVAR is chosen because to answer the research question requires precision, even at the cost of stronger modelling assumptions.

3.1.1 Lag Length Selection and applied tests

Lag length, essentially the delay between cause and effect, matters for the results. Too few lags risks omitting relevant dynamics and inducing residual autocorrelation; too many lags consume degrees of freedom, a particular concern given the short post-2020 subsample. Lag length is determined using the Bayesian Information Criterion (BIC), which is preferred for its stronger consistency properties in finite samples (Schwarz, 1978). Where BIC and the Akaike Information Criterion (AIC) conflict, the Likelihood Ratio (LR) test is used as a decider. The final specification is validated by testing for residual autocorrelation using the Portmanteau test.

3.2 Data Description and Preparation

3.2.1 Sample and Variable Definitions

The dataset consists of monthly observations for the US and the UK from March 1996 to December 2025 ($T \approx 358$). This timeframe captures the Great Moderation (~ 1985 -2007), the Global Financial Crisis (2008-2010), and the post-2020 structural break identified by Reis (2022). US data is sourced from FRED and the Federal Reserve Bank of Cleveland; UK data is sourced from the Office for National Statistics and the Bank of England.

The four-variable system for inflation, inflation expectations, policy rate and 10-year bond yield ($y_t = [\pi_t, e_t, i_t, r_t^{10Y}]$) is constructed from the variables defined in Table 1.

Table 1: Variable definitions

Variable	Proxy	Purpose
Inflation (π)	YoY % change in CPI (US) / CPIH (UK)	Measures nominal price pressure
Long-run inflation expectations (e)	Cleveland Fed 10-year expected inflation (US) / BoE implied inflation from Index-linked gilt spot curve (UK)	Market-implied measure of the inflation anchor
Policy rate (i)	Federal funds rate (US) / Bank rate (UK)	The policy rate is the interest rate set by the central bank to influence borrowing costs across the economy.
10Y Bond yield (r^{10Y})	10-year nominal government bond yield	Long end of the term structure; includes both expectations and risk premium

Including 10-year expected inflation as a separate variable (e), rather than an output measure such as Industrial Production, is a deliberate modelling choice. The central research question is whether the inflation anchor held or eroded after 2020, including expectations directly in the VAR means the impulse responses show how the anchor itself responds to a supply shock, something an output variable cannot provide. The Cleveland Fed model for the US and the Bank of England's index-linked gilt spot curve for the UK are used as the expectations proxies. Both are standard market-implied measures in the macro-finance literature (Ang et al., 2007; Barr and Campbell, 1997).

The variable ordering in the VAR is as follows: inflation, expectations, policy rate, bond yield. While the SVAR is identified through sign restrictions rather than a Cholesky decomposition (Section 3.3), the ordering remains relevant for the Cholesky factor (P) used as the starting point for the sign restriction rotation algorithm. The chosen ordering reflects the assumption that inflation is the most exogenous variable in the system at monthly frequency, while bond yields are the most responsive to contemporaneous information.

3.2.2 Stationarity, Levels, and Structural Breaks

The VAR is estimated in levels because differencing would throw away the long-run relationships between inflation, rates, and yields that the model needs to capture (Sims et al., 1990). As Sims et al. (1990) demonstrate, estimation in levels produces consistent estimates of

the VAR coefficients regardless of the integration and co-integration properties of the data, provided the lag length is sufficient.

Given the 30-year span, the sample is likely to contain structural breaks. Zivot-Andrews (1992) tests are applied to inflation for each country to identify whether and when a structural break occurred. These tests provide formal evidence for the existence and timing of the post-2020 regime shift, and they justify the sub-sample split conducted in the empirical analysis (Section 4.3).

3.2.3 Data Considerations at the Zero Lower Bound

Between 2009 and 2022, policy rates sat at or near zero. The observed rate therefore understates the true stance of policy, since Quantitative Easing (QE), the central bank purchasing government bonds to lower long-term interest rates, provided additional accommodation (Wu and Xia, 2016). QE also compressed long-term yields independently of expectations (Joyce et al., 2011). Both effects may contaminate the VAR during this window. The subsample analysis in Section 4.3 provides partial mitigation; the baseline model does not adjust for QE directly, as discussed in Section 3.7.

3.3 Identification Strategy: Sign Restrictions

The central empirical question of this dissertation, whether domestic anchoring or global contagion dominated UK bond markets after 2020, requires an identification strategy that distinguishes shocks by their macroeconomic origin. Evidence of anchoring would appear as a smaller and shorter-lived response of UK 10-year yields and expectations to a supply inflation shock, relative to the US. Evidence of global contagion would appear as a convergence of those responses, regardless of institutional differences.

To achieve this, recursive (Cholesky) identification is not the most effective choice. As Sims (1992) demonstrated, recursive SVARs generates the “Price Puzzle”, a contractionary monetary shock appears to raise inflation in the short run because the Cholesky ordering does not allow the policy rate to respond to expected future inflation. Since this analysis centres on the bond market response to inflation shocks, an identification scheme with this flaw would compromise the core results.

Instead, this study employs sign restrictions following Uhlig (2005), operationalising the demand/supply shock taxonomy of Cieřlak and Pflueger (2023). The structural impact matrix, B , is constructed by drawing random orthogonal matrices Q via QR decomposition of a random normal matrix, and computing $B = PQ$ where P is the Cholesky factor of the reduced-form covariance matrix. Column permutations and sign flips are applied to each draw to maximise the acceptance rate. A draw is retained if it satisfies the restrictions in Table 2 on the impact matrix (horizon 0). The algorithm generates 500 accepted draws for each country model.

Table 2: Sign Restriction Scheme (Impact Restrictions) following Uhlig (2005)

Shock Type	Inflation (π)	Expectations (e)	Policy Rate (i)	10Y Yield (r^{10y})
Supply (“bad” inflation)	+	unrestricted	unrestricted	+
Demand (“good” inflation)	+	+	+	+
Monetary policy	–	unrestricted	+	–

The supply shock is identified by a simultaneous rise in inflation and 10-year bond yields on impact, with expectations and the policy rate left unrestricted. The expectations variable is left unrestricted under the supply shock because its response is the object of interest: if the anchor holds, expectations should not respond persistently; if it fails, they should. Imposing a sign on expectations would predetermine the answer to the research question. The demand shock requires all four variables to rise on impact, consistent with a positive aggregate demand stimulus. The monetary policy shock is identified by a rise in the policy rate accompanied by a fall in both inflation and bond yields, with expectations unrestricted.

Sign restrictions produce set-identification rather than point-identification. The restrictions are satisfied by a set of admissible structural models, not a single unique model. The implications of this for reporting and inference are addressed in Section 3.4.

3.4 Inference: Impulse Responses and Variance Decompositions

Impulse Response Functions (IRFs) trace the dynamic path of expectations and 10-year yields following each identified shock, directly operationalising the anchoring hypothesis. Forecast Error Variance Decompositions (FEVDs) decompose which shock types are the dominant

drivers of expectations and yield variance, providing a quantitative basis for distinguishing between domestic and global explanations of bond market behaviour.

Sign restrictions don't give a single model. They produce hundreds of outcomes that all satisfy the constraints. To select a representative result, the Median Target Response (MTR) method of Fry and Pagan (2011) is employed in this dissertation. The MTR selects the single draw from the 500 accepted draws whose FEVD is simultaneously closest to the median FEVD across all variables and horizons, measured by the sum of squared deviations. This is preferable to the conventional element-wise median, which reports responses that may not correspond to any single admissible model and can therefore produce internally inconsistent inference. The MTR-selected draw is then used to compute both the reported FEVD and the structural impulse response functions.

If supply inflation shocks account for a substantially larger share of 10-year expectations variance in the US than in the UK, this constitutes evidence that domestic anchoring has insulated the UK's inflation expectations. If the variance shares converge, or if supply shocks explain a larger proportion of UK expectations variance post-2020, this supports the hypothesis that the Bank of England's credibility eroded during the inflation surge. Following Ehrmann and Fratzscher (2005), a difference of 10 percentage points in supply-shock variance shares between the two countries is adopted as the threshold for economically meaningful divergence.

3.5 Comparative Framework

The comparison between the US and UK uses a parallel estimation design. Two country-specific SVARs are estimated separately over the full sample (1996–2025) using identical variable definitions, lag selection procedures, and sign restriction schemes. This ensures that any differences in impulse responses reflect genuine country-level variation in shock transmission rather than specification differences.

IRFs for each shock type are plotted side-by-side, allowing visual comparison of the magnitude, persistence, and sign of the response across countries. Second, the sample is split into a pre-COVID baseline regime (March 1996–December 2019) and a post-COVID stress-test regime (January 2020–December 2025). The sign-restricted SVAR is re-estimated separately for each subsample, applying the same restriction scheme to ensure cross-regime comparability. BIC is used to select the lag length for the pre-COVID subsample; the post-

COVID subsample, with approximately 62 observations, uses a fixed lag length of 2 to preserve degrees of freedom. Comparing the FEVD results across regimes provides a direct test of whether the anchoring benefits documented by Gürkaynak et al. (2005) have persisted or eroded during the largest inflation shock in four decades.

3.6 Robustness Checks

In order to check the robustness of the model, both reduced-form Cholesky FEVDs and sign-restricted FEVDs are calculated. If the qualitative pattern of transatlantic divergence is consistent across both identification approaches, one recursive, one sign-restricted, this provides reassurance that the findings are not an artefact of a particular identification scheme. Second, the full-sample results are compared to the pre-COVID and post-COVID subsample estimations to directly isolate the regime dynamics and test whether the transatlantic divergence identified in the full sample holds, intensifies, or reverses under the high-inflation regime.

3.7 Limitations

The parallel country-specific SVAR design does not explicitly model contemporaneous spillover channels between the US and UK. Within-month transmission, for instance, a surprise Federal Reserve announcement affecting gilt yields on the same day, is captured only through the lagged VAR structure, which may underestimate the speed and magnitude of global contagion effects. A joint two-country SVAR with cross-equation restrictions would address this but at the cost of substantially increased dimensionality and identification complexity.

Sign restrictions are imposed on the impact matrix only (horizon 0) rather than over multiple horizons. While multi-horizon restrictions (e.g., imposing signs for four quarters as in Uhlig, 2005) would further sharpen identification, they are computationally demanding and risk over-constraining the model with a limited post-2020 subsample. The impact-only approach is conservative: it allows the data to determine the dynamic response at all subsequent horizons, reducing the risk that reported impulse responses are driven by the restriction design rather than the data.

The baseline model uses the observed policy rate throughout the sample without adjusting for the zero lower bound (ZLB) or the effects of Quantitative Easing (QE). During the 2009–2022

period, the policy rate understates the effective stance of monetary policy, and QE-driven yield compression may contaminate the bond yield variable. The subsample analysis provides partial mitigation by isolating the pre-ZLB and post-ZLB periods, but a full treatment would require replacing the policy rate with a shadow rate measure (Wu and Xia, 2016) and adjusting yields for the term premium.

Finally, the four-variable system does not include a direct measure of real economic activity, such as Industrial Production. The inclusion of inflation expectations in place of an output variable means that the supply and demand shocks are identified without an explicit output restriction. A five-variable system including both expectations and output would allow for an improved identification scheme but would increase the dimensionality of the sign restriction problem and reduce the acceptance rate, particularly in the short post-2020 subsample.

Chapter 4: Empirical Results

This chapter presents the empirical findings of the sign-restricted Structural Vector Autoregression (SVAR) analysis. Structural identification follows the sign restriction scheme outlined in Chapter 3 (Section 3.3), grounded in the demand and supply shock taxonomy of Cieřlak and Pflueger (2023). The Median Target Response (MTR) method of Fry and Pagan (2011) is employed throughout to select the most representative model.

Section 4.1 details the model diagnostics and structural break testing. Section 4.2 then analyses the full-sample (1996–2025) transmission of supply inflation shocks to market expectations and 10-year bond yields. Section 4.3 conducts a regime-shift analysis, explicitly testing the "bending anchor" hypothesis by comparing the pre-COVID (1996–2019) monetary regime against the post-COVID (2020–2025) supply shock environment.

4.1 Model Diagnostics and Specification

Zivot-Andrews unit root tests were employed to identify structural breaks within the inflation series. As detailed in Table 3, the test statistics identified highly significant structural breaks at observation 304 (mid-2021) for the United States, with a test statistic of -5.69 exceeding the 1% critical value of -5.57, and at observation 309 (late-2021) for the United Kingdom, with a test statistic of -5.16 exceeding the 5% critical value of -5.08. The coincident break in both countries suggests the inflation regime shift was a shared global phenomenon rather than a country-specific shock, providing preliminary evidence that global contagion was driving the inflation surge both central banks faced. These breakpoints align precisely with the onset of the post-COVID global supply chain disruptions and the subsequent energy price crisis, supporting the regime-shift subsample analysis conducted in Section 4.3.

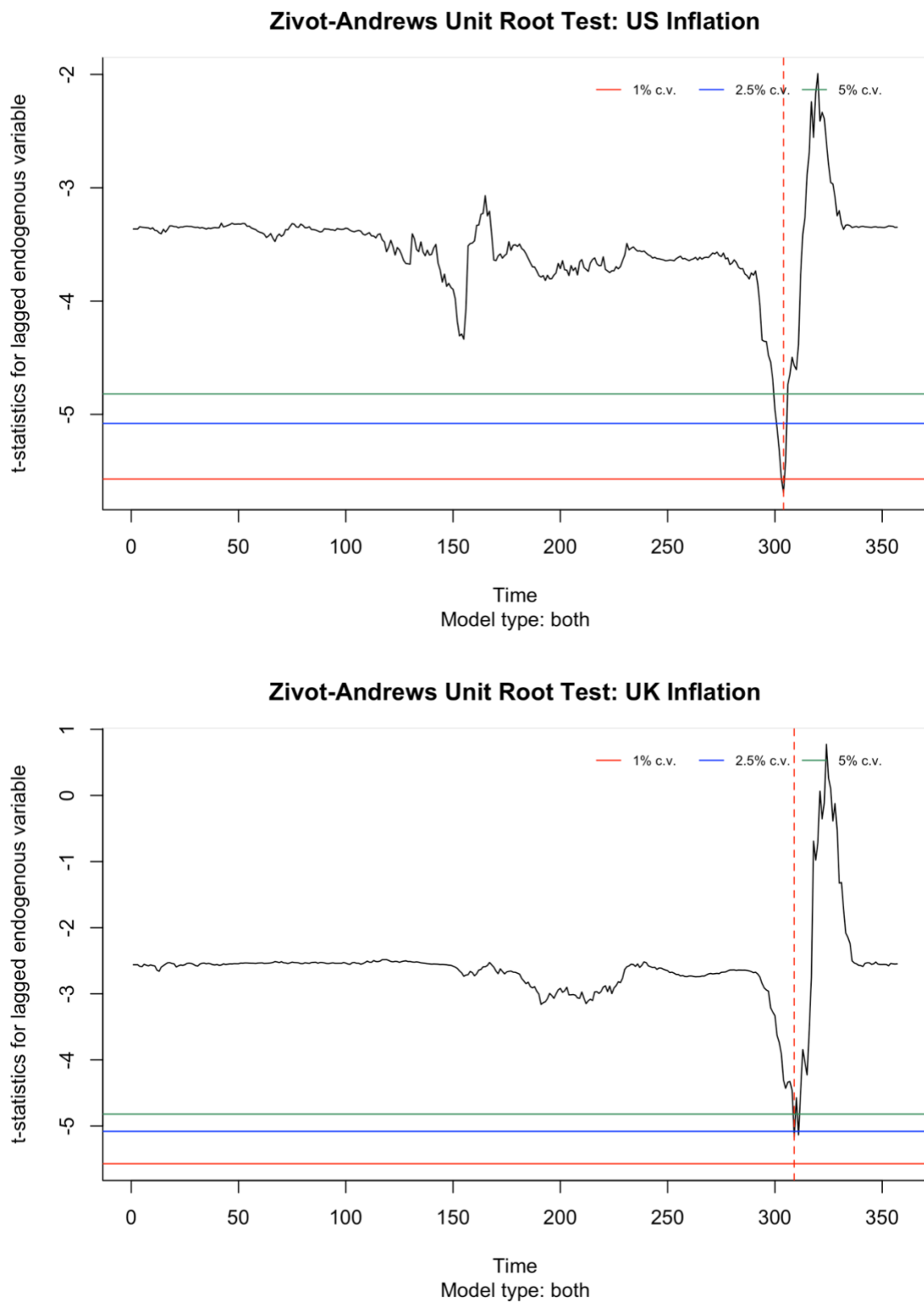
Lag lengths of 5 and 3 months were selected for the US and UK models respectively, based on the Bayesian Information Criterion (BIC). While asymptotic Portmanteau tests indicate the presence of residual serial correlation, a standard feature of highly persistent macro-finance data, preference was given to the BIC to maintain model parsimony and preserve degrees of freedom for the shorter post-2020 subsample estimations, following Stock and Watson (2007). The sign restriction algorithm achieved acceptance rates of 2.97% and 5.02% for the US and UK models respectively across 500 accepted draws, consistent with the expected rates for four-variable SVARs under multi-shock restrictions (Uhlig, 2005).

Table 3: Model Diagnostics and Structural Breaks

Metric	US Model	UK Model
Observations	358	358
Variables	4	4
Optimal Lags (BIC)	5 months	3 months
Zivot-Andrews Breakpoint (Inflation)	Obs. 304 (Mid-2021)	Obs. 309 (Late-2021)
ZA Test Statistic	−5.69***	−5.16**
ZA Critical Values (1% / 5% / 10%)	−5.57 / −5.08 / −4.82	−5.57 / −5.08 / −4.82
Serial Correlation (Portmanteau p-value)	< 0.001	< 0.001
Sign Restriction Acceptance Rate	2.97%	5.02%
Accepted Draws	500	500

*Notes: *** significant at 1%, ** significant at 5%. Zivot-Andrews test allows for endogenous break in both intercept and trend (model = “both”). Portmanteau test conducted with 16 lags. Sign restrictions imposed on impact matrix (horizon 0) with column permutations and sign flips. Random seed: 202213724.*

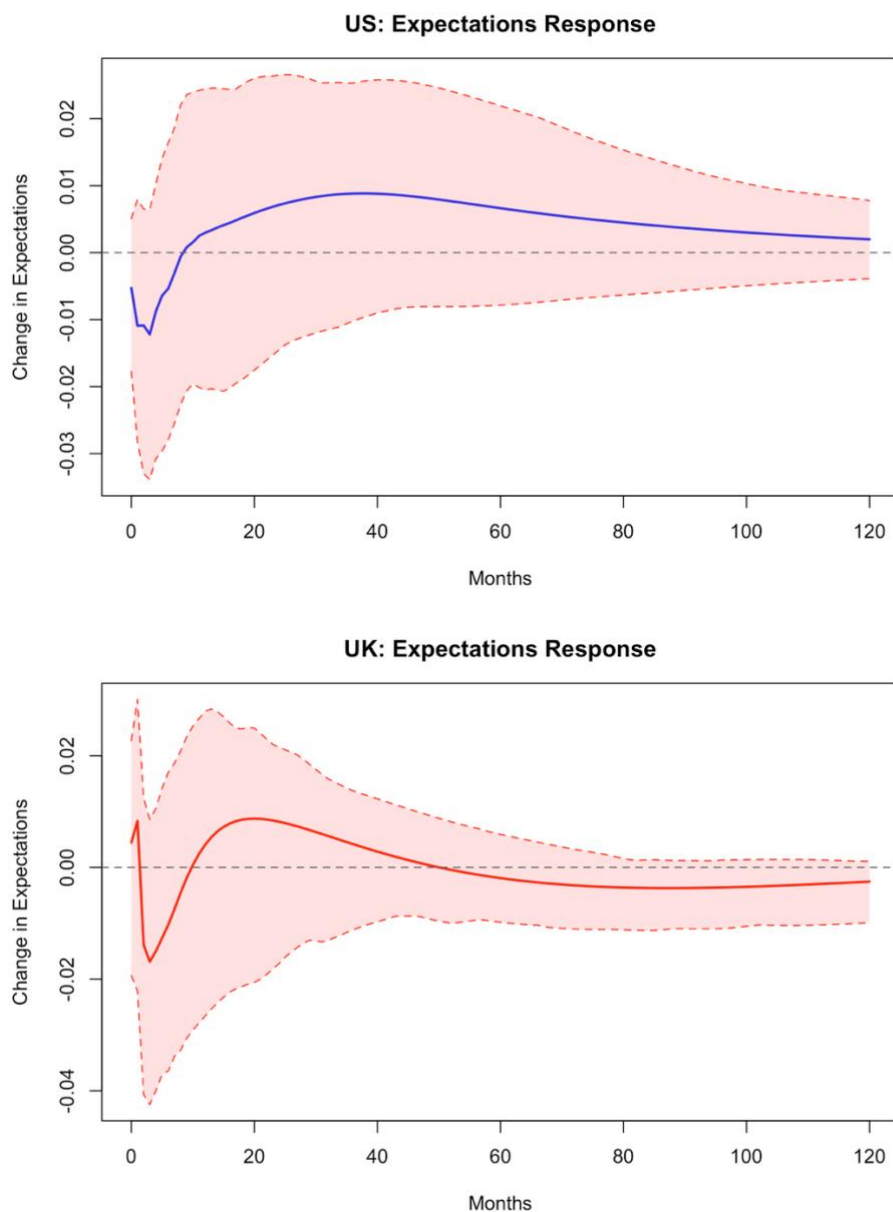
Figure 3: Zivot-Andrews Unit Root Test with Endogenous Structural Break, US and UK Inflation.



4.2 The Full-Sample Analysis: The Transatlantic Divergence (1996–2025)

To evaluate the baseline credibility of the Federal Reserve and the Bank of England over the full 30-year sample timeframe, the sign-restricted SVAR was estimated using the MTR-selected draw from 500 accepted identification schemes. Figure 4 illustrates the bootstrapped Impulse Response Functions (IRFs) mapping the dynamic reaction of 10-year implied inflation expectations to a one standard deviation supply shock, as defined in the identification scheme of Section 3.3.

Figure 4: Impulse Response Functions — 10-Year Implied Inflation Expectations to a Supply Shock, US and UK (Full Sample, 1996–2025).



The IRFs demonstrate that long-term expectations in both jurisdictions remain broadly anchored following the supply shock, with responses converging back toward the zero baseline over the medium term (Figure 4). This visual evidence is supported by the 120-month FEVD reported in Table 4. Critically, a meaningful transatlantic gap is visible in the full-sample period.

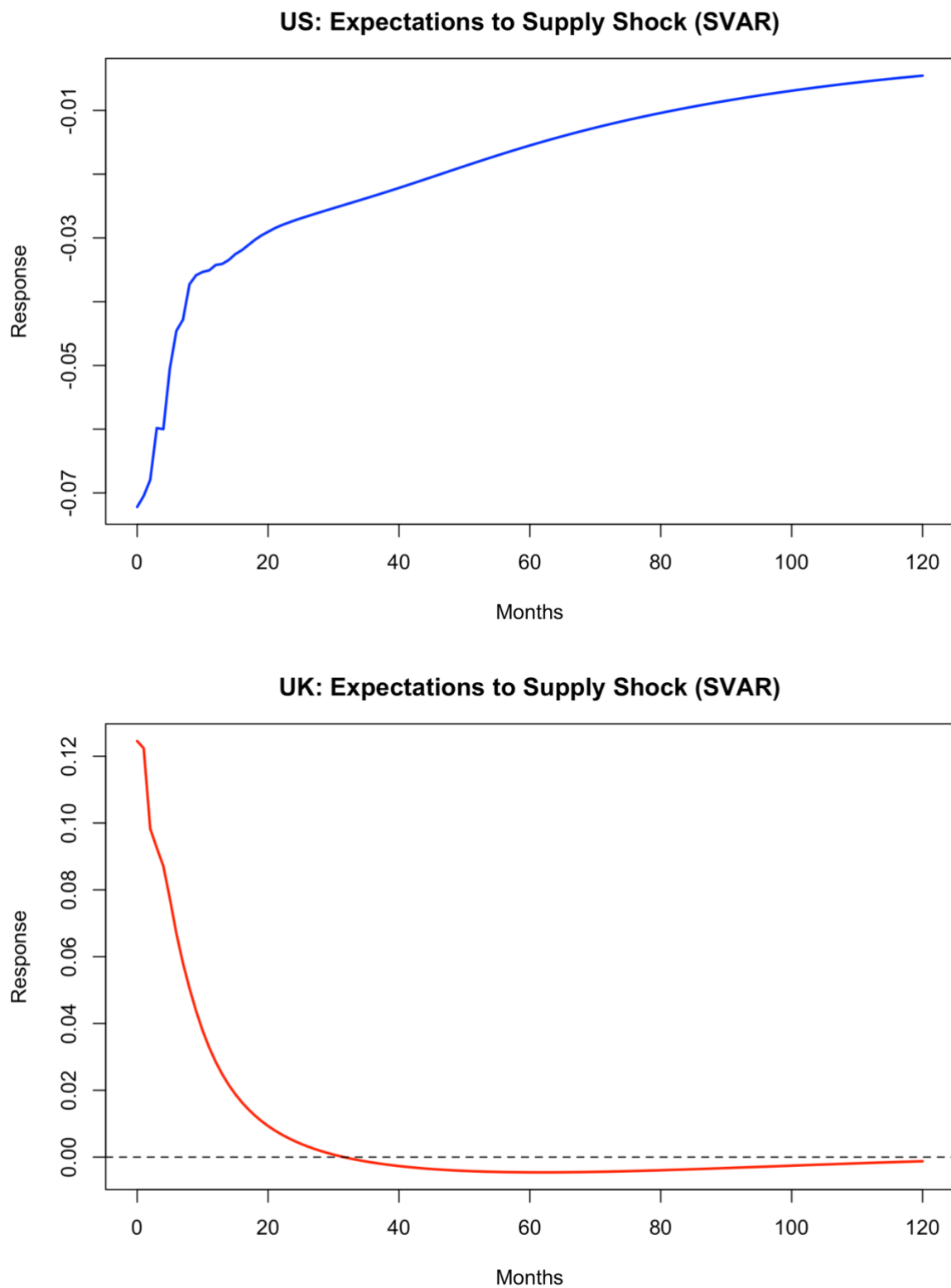
Table 4: Full Sample Forecast Error Variance Decomposition (FEVD) – Supply Shock Share (1996–2025)

Variable Responding to Supply Shock	US	UK
Long-Term Expectations (The Anchor)	6.24%	6.67%
10-Year Bond Yields (Risk Premium)*	2.95%	10.17%

**Bond yield figures derived from reduced-form Cholesky FEVD for comparative purposes.*

The variance decomposition provides no support for the "bending anchor" hypothesis over the full sample. Supply shocks explain 6.24% of the variance in US long-term expectations compared to 6.67% in the UK over the 120-month horizon. While both figures confirm a high degree of institutional credibility, the overwhelming majority of expectation variance is driven by own-persistence rather than inflation shocks, the UK's elevated sensitivity is economically meaningful. The gap suggests UK inflation expectations are more responsive to supply-side pressure than those in the US: not dramatically so, but enough that investors price a slightly higher risk of the Bank of England falling behind the curve.

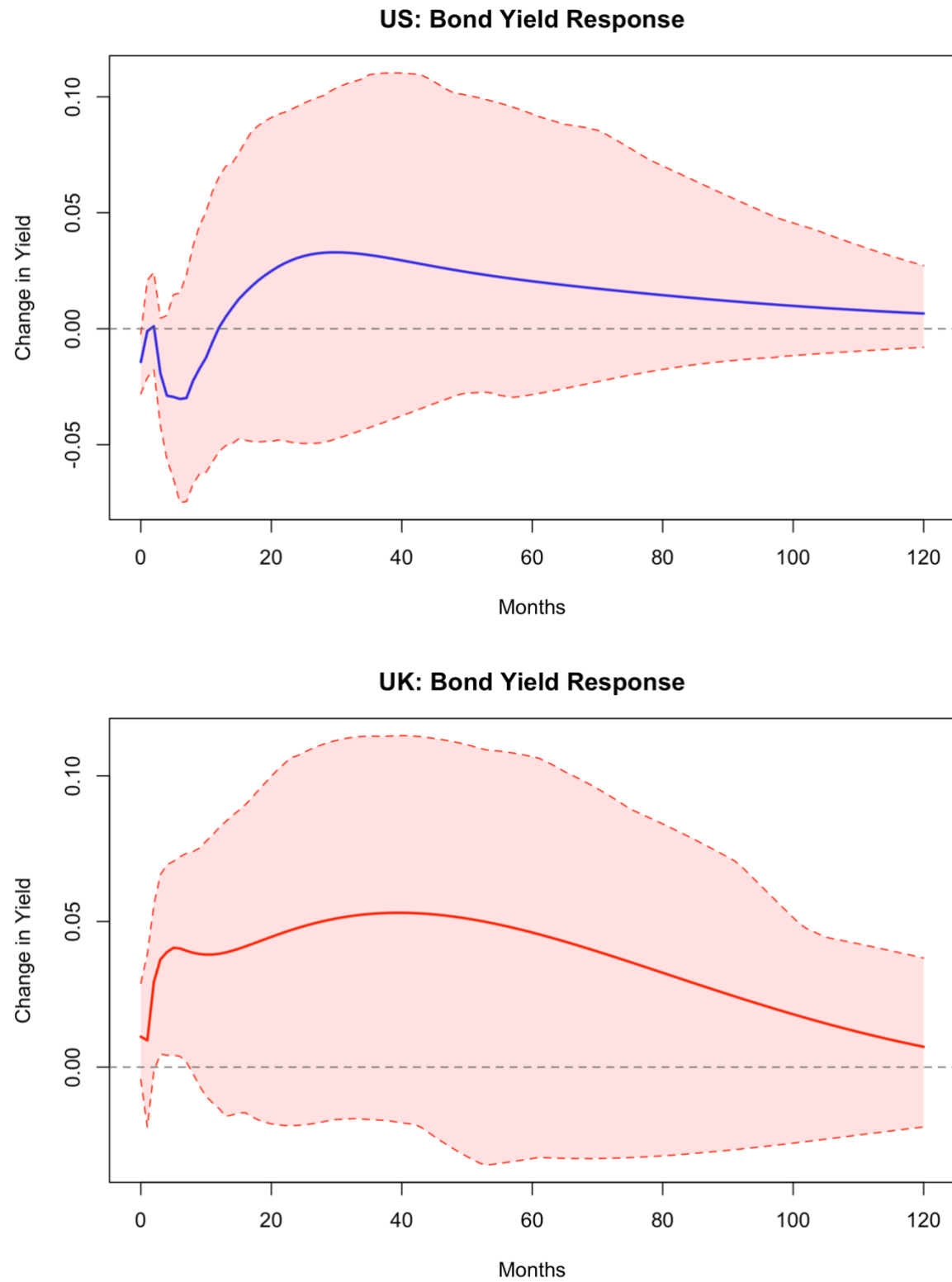
Figure 5: Comparative Impulse Response – 10-Year Inflation Expectations to a Supply Shock, US vs UK (Full Sample, MTR-Selected Draw).



A more pronounced transatlantic divergence emerges in the sovereign bond markets. As shown in Table 4, supply shocks explain 10.17% of the variance in UK gilt yields compared to just 2.95% for US Treasuries. Because this excess volatility in UK bond yields cannot be attributed to shifts in long-term inflation expectations alone, UK expectations and US expectations both remain largely anchored, it must reflect a differential Inflation Risk Premium embedded in gilt markets. Market participants evidently trust the Bank of England's long-run intentions, yet demand significantly higher yield compensation for holding UK sovereign debt during inflationary supply shock episodes. The distinction is important: the UK's credibility is intact at the level of belief but carries a higher price tag at the level of risk. This is consistent with the Global Financial Cycle hypothesis of Miranda-Agrippino and Rey (2020): while domestic anchoring insulates expectations, UK gilt markets remain exposed to the global risk-off dynamics amplified by supply shocks (Pinter, 2023), in a way that the US Treasury market, insulated by its safe-haven status, does not.

These full-sample findings provide an important baseline: both anchors hold over the long run, but the UK exhibits a structurally higher sensitivity to supply shocks at both the expectations and bond yield level. The question addressed in Section 4.3 is whether this gap widened, narrowed, or inverted following the COVID-19 supply shock stress test.

Figure 6: Impulse Response Functions – 10-Year Government Bond Yields to a Supply Shock, US and UK (Full Sample, 1996–2025).



4.3 Regime Shifts: The COVID-19 Supply Shock Stress Test

If the Bank of England's inflation-targeting framework successfully insulated the UK, we should see UK gilt yields respond less strongly and recover faster than US Treasuries after a supply shock. If the Global Financial Cycle dominated, we should see both countries respond similarly — the institutional differences becoming irrelevant when a sufficiently large global shock hits. The results below point clearly in one direction for the post-2021 period. While the full-sample analysis establishes the baseline for transatlantic credibility, it obscures the intense structural breaks identified during the 2021 inflationary surge (Section 4.1). To explicitly test whether the post-2020 environment fundamentally altered central bank credibility the data was partitioned into a pre-COVID baseline regime (March 1996 – December 2019) and a post-COVID stress-test regime (January 2020 – December 2025). The MTR-identified sign-restricted SVAR was re-estimated separately for each subsample, applying the same restriction scheme to ensure cross-regime comparability.

Table 5 presents the supply shock variance decomposition for both regimes, revealing a stark and theoretically significant shift in transatlantic credibility dynamics.

Table 5: Credibility Stress Test – Supply Shock FEVD by Regime

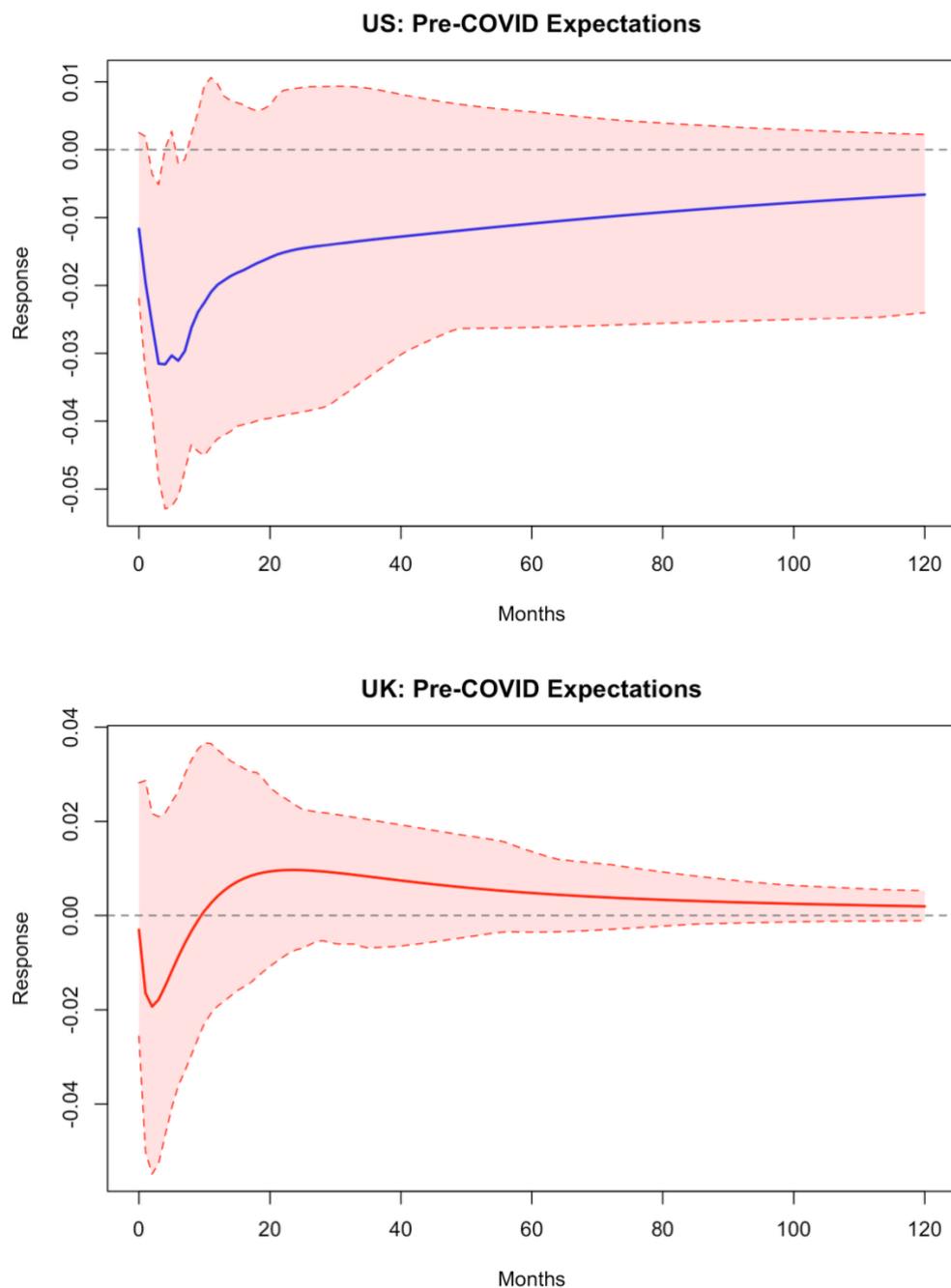
Regime	US Anchor	UK Anchor
Pre-COVID (1996–2019)	11.91%	7.71%
COVID-Era (2020–2025)	12.60%	12.73%

4.3.1 The Pre-COVID Baseline: Bank of England's Advantage

Prior to 2020, the Bank of England's inflation anchor was meaningfully stronger than that of the Federal Reserve (Table 5). During the 1996–2019 regime, supply shocks drove 11.91% of the variance in US long-term expectations, compared to just 7.71% in the UK (Table 5). This differential, while modest in absolute terms, is directionally consistent with the thesis of Gürkaynak et al. (2005). They demonstrated using high-frequency event studies that the Bank of England's explicit inflation-targeting mandate, adopted in 1992, anchored UK long-term forward rates more effectively than the Federal Reserve's dual-mandate framework, which historically allowed for greater policy ambiguity.

The pre-COVID IRFs reinforce this finding. The UK anchor's response to a supply shock decays more rapidly toward zero than the US equivalent, indicating that financial markets treated supply-driven inflationary episodes as transitory relative to the Bank of England's 2% target (Figure 7). Prior to the pandemic, UK sovereign debt therefore benefitted from a better tier of institutional credibility.

Figure 7: Impulse Response Functions – 10-Year Inflation Expectations to a Supply Shock, US and UK (Pre-COVID Subsample, 1996–2019).



4.3.2 The Post-COVID Reversal: Credibility Convergence

The onset of the 2020 global supply chain disruption and the subsequent 2022 energy price crisis following Russia's invasion of Ukraine served as a useful stress test for this institutional trust. The results of the regime-shift analysis show the transatlantic credibility differential converged sharply during this period (Figure 8).

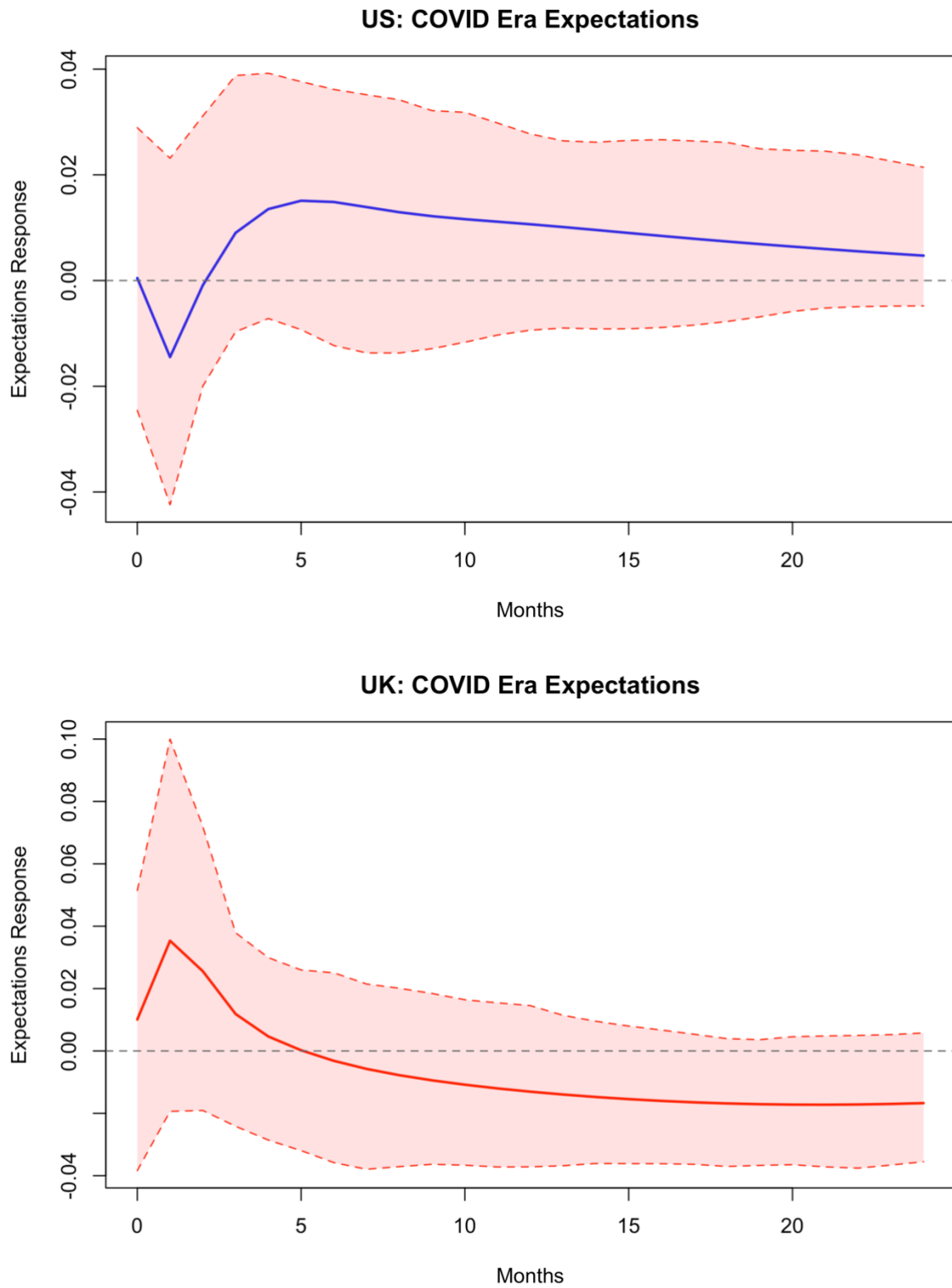
Despite consumer price inflation reaching a four-decade high of 9.1% in June 2022, the sensitivity of US long-term expectations to supply shocks held relatively stable from the pre-COVID baseline, going from 11.91% compared to 12.60%. This stability strongly suggests that financial markets trusted the Federal Reserve's aggressive, front-loaded rate-hiking cycle, the fastest tightening cycle since the 1980s Volcker era, as a credible commitment to restoring price stability. The US anchor absorbed a historically severe supply shock with minimal change.

The UK experience was very different. The sensitivity of UK long-term expectations to supply shocks rose sharply during the COVID-era regime, rising from 7.71% pre-COVID to 12.73%, an increase of 5.02 percentage points. Markets began to price a substantially elevated probability that supply-driven inflation would become entrenched in UK long-run expectations, directly eroding the Bank of England's credibility advantage.

This divergence reflects the UK's unique macroeconomic vulnerabilities during this period. As a smaller, highly open economy, the UK was disproportionately exposed to imported energy price inflation following the Ukraine conflict, with natural gas prices amplifying domestic price stickiness to a degree not experienced in the more energy-independent US economy. Post-Brexit supply chain frictions further compounded inflationary pressures, effectively making the Bank of England's mandate harder to achieve than the Federal Reserve's under otherwise comparable global conditions.

Applying the Bernanke-Blanchard framework to the UK specifically, Haskel (2024) finds that tight labour market conditions entering the pandemic amplified second-round effects on domestic inflation. This provides a microeconomic channel through which the UK's institutional anchor faced greater stress than its US counterpart.

Figure 8: Impulse Response Functions – 10-Year Inflation Expectations to a Supply Shock, US and UK (Post-COVID Subsample, 2020–2025).



4.3.3 The "*Bending Anchor*" Hypothesis

The post-2020 credibility change is real and statistically meaningful. The UK lost its pre-COVID advantage over the US, with supply shock sensitivity rising by 5.02 percentage points compared to an essentially flat US trajectory. The COVID-19 supply shock stress test exposed structural vulnerabilities in the UK's inflation anchoring capacity that the Federal Reserve's framework successfully withstood.

At 12.73%, supply shocks drive a substantially larger share of UK expectation variance than in the pre-COVID period. Still 87.27% of variance comes from other factors. Furthermore, this figure, while elevated, does not approach the threshold at which expectations might be described as fully unanchored from the 2% target. The supply shock variance share remains within the range of what macro-finance literature documents for economies with functioning inflation-targeting frameworks under stress (Gürkaynak et al., 2005; Reis, 2022).

The Bank of England anchor bent rather than broke. This aligns with Bundick et al. (2024), who find that while long-run credibility remained intact during the pandemic, advanced economies experienced significant short-to-medium term drift in expectations. Bailey (2024) similarly acknowledged that the Ukraine energy shock substantially increased the risk of de-anchoring but concluded that expectations appeared better anchored than in previous episodes, attributing this to independent central banks with clear mandates. The Bank of England suffered a structural credibility blow that the Federal Reserve avoided, forcing markets to reassess the relative safety of UK sovereign assets. The 68th percentile ranges across all 500 accepted draws (Appendix Table A6) confirm that while credible intervals are wide (a standard feature of set-identified SVARs) the directional shift in UK sensitivity is consistent across the majority of draws. Combined with the elevated inflation risk premium in UK gilt yields (Section 4.2: 10.17% of bond yield variance vs 2.95% for US Treasuries), this regime shift explains the large transatlantic divergence in sovereign bond market dynamics observed post-2020. The following chapter discusses implications for the broader literature on domestic anchoring versus global financial cycle contagion.

Chapter 5: Conclusion

This dissertation set out to answer a specific question: ‘did the Bank of England’s inflation-targeting framework continue to anchor bond market expectations more effectively than the Federal Reserve’s dual mandate during the post-2020 inflation shock?’. The answer, based on a sign-restricted SVAR estimated for the period 1996–2025, is that it did not. The pre-existing credibility advantage documented in the literature eroded sharply during the very period when it was needed most.

The full-sample results confirm that, taken over three decades, both central banks maintained broadly credible inflation anchors. Supply shocks explained 6.24% of the variance in US long-term expectations and 6.67% in the UK, although interestingly these figures indicate that the vast majority of expectation dynamics are attributable to own-persistence rather than inflationary pressure. Markets, in aggregate, were confident that both institutions would deliver on their mandates. At the same time, a structural asymmetry was already visible in sovereign bond markets. Supply shocks explained 10.17% of UK gilt yield variance compared to just 2.95% for the US Treasury.

Before 2020, supply shocks drove 7.71% of UK expectation variance compared to 11.91% for the US. The Bank of England held the advantage. After 2020, UK sensitivity rose significantly to 12.73% while the US figure stayed flat at 12.60%.

Explicit inflation targeting does anchor expectations (e.g. Gürkaynak et al., 2005), but the insulation it provides is not unconditional. It held during the Great Moderation and through the Global Financial Crisis, but partially gave way under the compound stress of a global supply shock layered on top of country-specific structural weaknesses. Institutional frameworks matter, but they are not immune to sufficiently large real-world shocks.

The results are consistent with the Global Financial Cycle hypothesis of Miranda-Agrippino and Rey (2020), but only partially. US monetary policy clearly influenced global risk pricing, and the Federal Reserve’s credibility appeared to benefit from the dollar’s safe-haven status in ways the Bank of England could not replicate. Yet the divergence in outcomes was not purely global, it reflected genuine domestic factors, particularly the UK’s energy dependence and post-Brexit supply constraints. Domestic anchoring held up to a point, but country-specific vulnerabilities determined where that point was.

5.1 Limitations

Several limitations should be noted when interpreting these findings. The parallel country-specific SVAR design does not explicitly model contemporaneous spillovers between US and UK markets. Within-month transmission (a surprise Federal Reserve announcement moving gilt yields on the same day) is captured only through the lagged structure of the VAR, which likely understates the speed and magnitude of cross-border contagion. A joint two-country SVAR with cross-equation restrictions would address this, but at the cost of substantially increased dimensionality that places it beyond the scope of this study.

Sign restrictions produce set-identification rather than point-identification, meaning the results reflect a set of admissible models rather than a single unique solution. The Median Target Response method of Fry and Pagan (2011) mitigates this by selecting the most representative draw, the approach carries some inherent ambiguity. Restrictions are also imposed at impact only; multi-horizon restrictions would sharpen identification but risk over-constraining the model given the short post-2020 subsample.

The baseline model uses the observed policy rate without adjusting for the zero lower bound or the effects of Quantitative Easing (QE). During the 2009–2022 period, the policy rate understates the true stance of monetary policy, and QE-driven yield compression may contaminate the bond yield variable. The subsample split provides partial mitigation by isolating pre- and post-ZLB periods, but a full treatment would require a shadow rate measure (Wu and Xia, 2016).

Finally, the four-variable system does not include a direct measure of real economic activity. The inclusion of inflation expectations in place of an output variable was a deliberate choice, the anchor is what is tested in this thesis, but it means the supply-demand shock decomposition rests entirely on the co-movement of inflation, expectations, the policy rate, and bond yields, without an explicit output restriction.

5.2 Further Research

Incorporating a shadow rate would allow the monetary policy channel to be modelled more accurately through the ZLB period. Extending the framework to a joint two-country SVAR with explicit spillover channels would test the Global Financial Cycle hypothesis more directly. Adding an output variable would sharpen the supply-demand identification, though at the cost

of increased dimensionality. As the post-2020 sample lengthens beyond its current 62 observations, it will also become possible to test with greater statistical power whether the UK credibility gap narrows as the Bank of England's tightening cycle matures, or whether the credibility erosion documented here proves to be a lasting structural shift in transatlantic sovereign bond dynamics.

Word Count: 7993

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Appendices

Figure A1: Impulse Responses for Demand and Monetary Policy Shocks

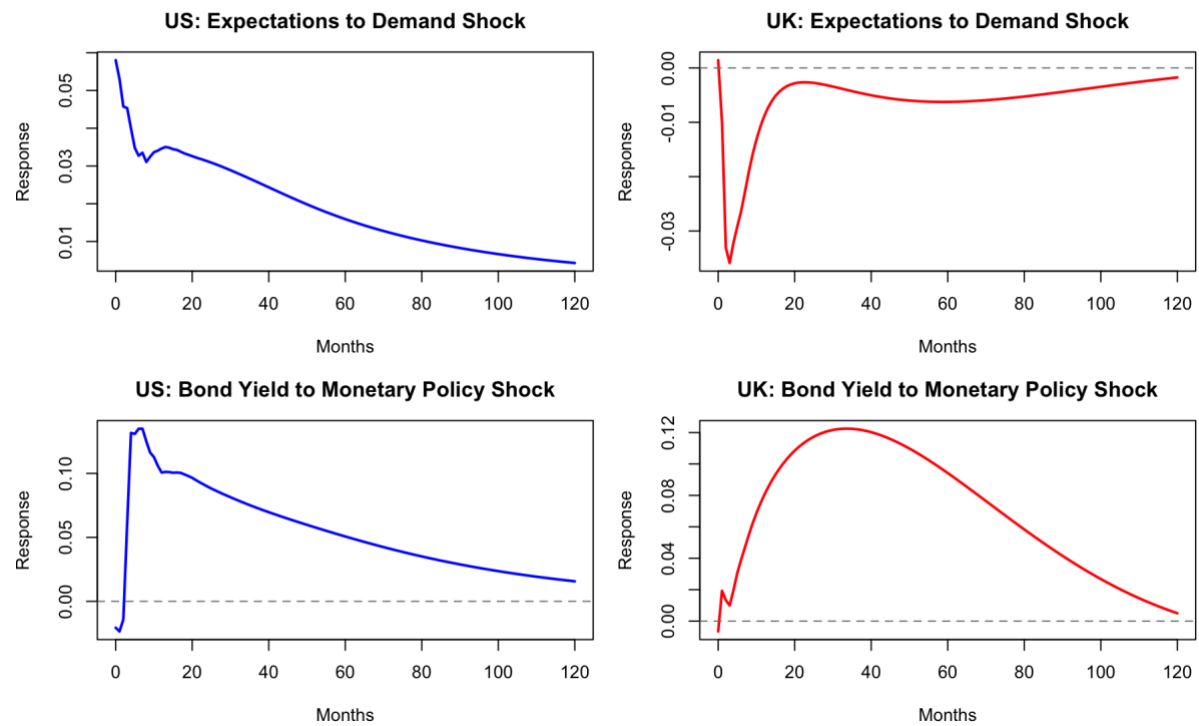


Figure A2: 68th and 90th Percentile Credible Sets for the Supply Shock Across all 500 Accepted Draws

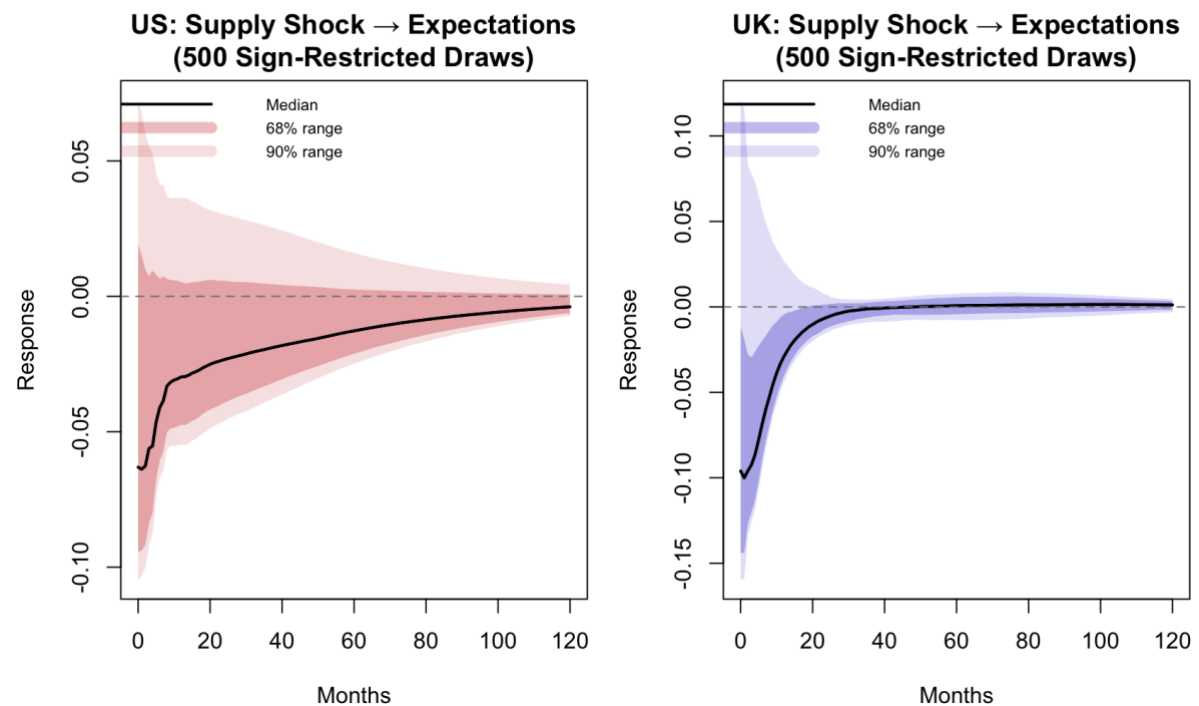


Table A1: Full Sample FEVD – Sign-Restricted, MTR-Selected Draw (1996–2025)*Forecast error variance decomposition at 120-month horizon. All values in percentages.****US Model***

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	6.24	26.60	2.98	64.19
Expectations	33.42	29.40	33.98	3.20
Policy Rate	19.49	13.18	46.07	21.25
Bond Yield	23.48	41.90	29.03	5.58

UK Model

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	6.67	36.85	9.05	47.43
Expectations	45.04	4.94	38.24	11.78
Policy Rate	7.15	11.16	74.68	7.01
Bond Yield	13.08	27.09	44.48	15.35

Notes: Sign restrictions follow Uhlig (2005). Supply shock: inflation (+), bond yield (+). Demand shock: all variables (+). Monetary policy shock: inflation (–), policy rate (+), bond yield (–). MTR draw selected per Fry and Pagan (2011).

Table A2: Pre-COVID FEVD – Sign-Restricted, MTR-Selected Draw (1996–2019)*Forecast error variance decomposition at 120-month horizon. All values in percentages.***US Model**

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	11.91	42.98	9.85	35.25
Expectations	23.80	28.77	3.21	44.23
Policy Rate	19.84	43.04	11.81	25.31
Bond Yield	15.50	26.77	5.51	52.22

UK Model

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	7.71	62.20	3.90	26.18
Expectations	43.26	1.30	51.40	4.04
Policy Rate	6.57	6.37	22.85	64.20
Bond Yield	9.01	10.57	11.51	68.92

Table A3: COVID-Era FEVD – Sign-Restricted, MTR-Selected Draw (2020–2025)*Forecast error variance decomposition at 24-month horizon. All values in percentages.***US Model**

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	12.51	13.00	47.95	26.54
Expectations	56.94	16.08	14.50	12.48
Policy Rate	12.00	44.81	27.29	15.90
Bond Yield	23.39	47.70	9.39	19.52

UK Model

Variable	Supply	Demand	Monetary Policy	Residual
Inflation	12.82	24.76	48.29	14.14
Expectations	32.78	10.20	9.53	47.49
Policy Rate	21.67	41.19	27.80	9.34
Bond Yield	19.77	33.60	19.01	27.62

Table A4: Robustness Check – Cholesky vs Sign-Restricted FEVD

Supply shock share of long-term inflation expectations variance (%). Cholesky uses recursive ordering; sign restrictions follow Uhlig (2005) with MTR selection.

Sample Period	Country	Cholesky (%)	Sign-Restricted (%)	Horizon
Full Sample (1996–2025)	US	2.12	6.24	120m
	UK	1.71	6.67	120m
Pre-COVID (1996–2019)	US	8.07	11.91	120m
	UK	3.09	7.71	120m
COVID-Era (2020–2025)	US	5.33	12.60	24m
	UK	5.94	12.73	24m

Notes: Under both identification approaches, the UK exhibits higher supply-shock sensitivity than the US in the full sample and pre-COVID period. The post-COVID convergence pattern is also consistent across both methods. The qualitative findings of the main text are robust to the choice of identification scheme.

Table A5: Full Sample Cholesky FEVD (1996–2025)

Reduced-form Cholesky variance decomposition at 120-month horizon. Ordering: inflation, expectations, policy rate, bond yield. All values in percentages.

US Model

Variable	Inflation	Expectations	Policy Rate	Bond Yield
Inflation	74.72	15.23	1.36	8.68
Expectations	2.12	95.44	1.82	0.62
Policy Rate	6.92	59.98	23.71	9.39
Bond Yield	2.95	88.99	2.04	6.02

UK Model

Variable	Inflation	Expectations	Policy Rate	Bond Yield
Inflation	66.88	17.68	2.38	13.06
Expectations	1.71	87.51	2.27	8.51
Policy Rate	3.67	38.69	53.95	3.69
Bond Yield	10.17	49.89	13.85	26.09

Table A6: Set-Identification Uncertainty – Supply Shock Share of Expectations Variance

Distribution across 500 accepted sign-restricted draws. MTR-selected draw (reported in main text) is the single draw closest to the median FEVD across all variables and horizons (Fry and Pagan, 2011).

Sample	Country	Median (%)	16th Pctl (%)	84th Pctl (%)
Full Sample (1996–2025)	US	31.52	7.13	64.75
	UK	43.32	15.88	73.66
Pre-COVID (1996–2019)	US	25.46	5.23	52.93
	UK	40.75	16.79	75.39
COVID-Era (2020–2025)	US	51.19	25.16	72.37
	UK	33.53	10.45	59.26

Notes: Wide credible intervals are a standard feature of set-identified SVARs (Fry and Pagan, 2011). The MTR-selected draw reported in the main text falls within the interquartile range for all samples. The intervals confirm that while the point estimates should be interpreted with caution, the directional shift in UK sensitivity post-2020 is consistent across the majority of accepted draws.

Table A7: VAR Lag Selection Summary

Lag length selected by Bayesian Information Criterion (BIC/SC). COVID-era subsample uses fixed lag length of 2 to preserve degrees of freedom.

Model	US Lags (BIC)	UK Lags (BIC)
Full Sample (1996–2025)	5	3
Pre-COVID (1996–2019)	5	2
COVID-Era (2020–2025)	2 (fixed)	2 (fixed)

A8: Dissertation Code

```
# Dissertation Code
# SVAR analysis: inflation shocks & central bank credibility (US vs UK)
set.seed(202213724)

library(vars)
library(urca)
setwd("/Users/freddiefuller/Desktop/Dissertation")

data <- read.csv("/Users/freddiefuller/Desktop/Dissertation/US & UK Inflation .csv", stringsAsFactors=FALSE)
clean_data <- data[, 3:10]
clean_data <- as.data.frame(lapply(clean_data, as.numeric))
clean_data <- na.omit(clean_data)

# ordering: inflation, expectations, interest rate, bond yield
us_ts <- clean_data[, c(1, 4, 2, 3)]
colnames(us_ts) <- c("US_Inflation", "US_Expectations", "US_Interest", "US_Bond")
uk_ts <- clean_data[, c(5, 8, 6, 7)]
colnames(uk_ts) <- c("UK_Inflation", "UK_Expectations", "UK_Interest", "UK_Bond")
cat("Data loaded:", nrow(us_ts), "observations\n")

## Zivot-Andrews tests (stationarity with structural breaks)

za_us_inf <- ur.za(us_ts[, "US_Inflation"], model = "both", lag = 2)
print(summary(za_us_inf))
par(mar = c(5, 4, 4, 2))
plot(za_us_inf)
rect(par("usr")[1], par("usr")[4], par("usr")[2], par("usr")[4] + 1, col = "white", border = "white", xpd = TRUE)
title(main = "Zivot-Andrews Unit Root Test: US Inflation")

za_uk_inf <- ur.za(uk_ts[, "UK_Inflation"], model = "both", lag = 2)
print(summary(za_uk_inf))
par(mar = c(5, 4, 4, 2))
plot(za_uk_inf)
rect(par("usr")[1], par("usr")[4], par("usr")[2], par("usr")[4] + 1, col = "white", border = "white", xpd = TRUE)
title(main = "Zivot-Andrews Unit Root Test: UK Inflation")

## US VAR

cat("\n--- US VAR ---\n")
lag_select_us <- VARselect(us_ts, lag.max = 12, type = "const")
p_us <- as.numeric(lag_select_us$selection["SC(n)"]) # BIC
cat("US lags (BIC):", p_us, "\n")

var_us <- VAR(us_ts, p = p_us, type = "const")
serial_test_us <- serial.test(var_us, lags.pt = 16, type = "PT.asymptotic")
print(serial_test_us)

## UK VAR

cat("\n--- UK VAR ---\n")
lag_select_uk <- VARselect(uk_ts, lag.max = 12, type = "const")
p_uk <- as.numeric(lag_select_uk$selection["SC(n)"])
cat("UK lags (BIC):", p_uk, "\n")

var_uk <- VAR(uk_ts, p = p_uk, type = "const")
serial_test_uk <- serial.test(var_uk, lags.pt = 16, type = "PT.asymptotic")
print(serial_test_uk)

## Sign-restricted SVAR (Uhlig 2005; Cieslak & Pflueger 2023)
# MTR draw selection per Fry & Pagan (2011)
#
# Restrictions (impact, horizon 0):
#   Supply: infl +, bond +
#   Demand: infl +, exp +, int +, bond +
#   Mon pol: infl -, int +, bond -

# FEVD from a single B draw (supply shock share only)
get_fevd_from_B <- function(var_model, B_draw, n.ahead = 120) {
  A_list <- Acoef(var_model)
  p <- var_model$p
  k <- ncol(B_draw)

  A_comp <- matrix(0, k*p, k*p)
  for (i in 1:p) A_comp[1:k, ((i-1)*k+1):(i*k)] <- A_list[[i]]
  if (p > 1) A_comp[(k+1):(k*p), 1:(k*(p-1))] <- diag(k*(p-1))

  MA <- array(0, dim = c(k, k, n.ahead+1))
  MA[,1] <- diag(k)
  companion_power <- diag(k*p)
  for (h in 1:n.ahead) {
    companion_power <- companion_power %*% A_comp
    MA[,h+1] <- companion_power[1:k, 1:k]
  }
}
```

```

# Main sign-restriction sampler
get_svar_draws <- function(var_model, n_draws = 500) {
  Sigma <- summary(var_model)$covres
  P <- t(chol(Sigma))
  n <- nrow(Sigma)
  accepted <- list()
  attempts <- 0

  perms <- list(c(1,2,3,4), c(1,3,2,4), c(2,1,3,4), c(2,3,1,4),
               c(3,1,2,4), c(3,2,1,4))

  while (length(accepted) < n_draws) {
    attempts <- attempts + 1
    Q <- qr.Q(qr(matrix(rnorm(n^2), n, n)))
    B_raw <- P %*% Q

    found <- FALSE
    for (perm in perms) {
      B <- B_raw[, perm]
      for (s1 in c(1,-1)) for (s2 in c(1,-1)) for (s3 in c(1,-1)) {
        Bt <- B
        Bt[,1] <- Bt[,1] * s1
        Bt[,2] <- Bt[,2] * s2
        Bt[,3] <- Bt[,3] * s3

        supply_ok <- Bt[1,1] > 0 && Bt[4,1] > 0
        demand_ok <- Bt[1,2] > 0 && Bt[2,2] > 0 && Bt[3,2] > 0 && Bt[4,2] > 0
        monopol_ok <- Bt[1,3] < 0 && Bt[3,3] > 0 && Bt[4,3] < 0

        if (supply_ok && demand_ok && monopol_ok) {
          accepted[[length(accepted) + 1]] <- Bt
          found <- TRUE
          break
        }
      }
    }
    if (found) break
  }

  if (attempts > 200000) {
    cat("Stopped after 200k attempts, got", length(accepted), "draws\n")
    break
  }
}

cat("Acceptance rate:", round(length(accepted)/attempts * 100, 2), "%\n")
return(accepted)
}

# MTR selection: pick the draw closest to the median FEVD
select_mtr_draw <- function(draws, var_model, n.ahead = 120) {
  all_fevd <- lapply(draws, function(B) get_fevd_from_B(var_model, B, n.ahead))
  fevd_array <- simplify2array(all_fevd)
  median_fevd <- apply(fevd_array, 1:2, median)
  dists <- sapply(all_fevd, function(f) sum((f - median_fevd)^2))
  return(draws[[which.min(dists)]])
}

# structural IRFs from a given B matrix
get_structural_irf <- function(var_model, B_draw, n.ahead = 120) {
  A_list <- Acoef(var_model)
  p <- var_model$p; k <- ncol(B_draw)
  A_comp <- matrix(0, k*p, k*p)
  for (i in 1:p) A_comp[1:k, ((i-1)*k+1):(i*k)] <- A_list[[i]]
  if (p > 1) A_comp[(k+1):(k*p), 1:(k*(p-1))] <- diag(k*(p-1))

  irf_array <- array(0, dim = c(k, k, n.ahead+1))
  irf_array[, , 1] <- B_draw
  companion_power <- diag(k*p)
  for (h in 1:n.ahead) {
    companion_power <- companion_power %*% A_comp
    irf_array[, , h+1] <- companion_power[1:k, 1:k] %*% B_draw
  }
  return(irf_array)
}

# Run sign restrictions
cat("\n--- US sign restrictions ---\n")
us_draws <- get_svar_draws(var_us, n_draws = 500)
B_us_mtr <- select_mtr_draw(us_draws, var_us, n.ahead = 120)

cat("\n--- UK sign restrictions ---\n")
uk_draws <- get_svar_draws(var_uk, n_draws = 500)
B_uk_mtr <- select_mtr_draw(uk_draws, var_uk, n.ahead = 120)

sirf_us <- get_structural_irf(var_us, B_us_mtr, n.ahead = 120)
sirf_uk <- get_structural_irf(var_uk, B_uk_mtr, n.ahead = 120)

par(mfrow = c(1, 2))
plot(0:120, sirf_us[2, 1, ], type = "l", main = "US: Expectations to Supply Shock (SVAR)",
     xlab = "Months", ylab = "Response", col = "blue")

```

```

abline(h = 0, lty = 2)
plot(0:120, sirf_uk[2, 1, ], type = "l", main = "UK: Expectations to Supply Shock (SVAR)",
     xlab = "Months", ylab = "Response", col = "red")
abline(h = 0, lty = 2)

# FEVD from MTR draw
us_fevd_mtr <- get_fevd_from_B(var_us, B_us_mtr, n.ahead = 120)
uk_fevd_mtr <- get_fevd_from_B(var_uk, B_uk_mtr, n.ahead = 120)

cat("\nFull sample FEVD - supply shock share at 120 months:\n")
cat("  US Expectations:", round(us_fevd_mtr[120,1]*100, 2), "%\n")
cat("  UK Expectations:", round(uk_fevd_mtr[120,1]*100, 2), "%\n")

## Cholesky IRFs
irf_us_exp <- irf(var_us, impulse = "US_Inflation", response = "US_Expectations",
                 n.ahead = 120, boot = TRUE, ci = 0.95)
irf_us_bond <- irf(var_us, impulse = "US_Inflation", response = "US_Bond",
                 n.ahead = 120, boot = TRUE, ci = 0.95)
irf_uk_exp <- irf(var_uk, impulse = "UK_Inflation", response = "UK_Expectations",
                 n.ahead = 120, boot = TRUE, ci = 0.95)
irf_uk_bond <- irf(var_uk, impulse = "UK_Inflation", response = "UK_Bond",
                 n.ahead = 120, boot = TRUE, ci = 0.95)

fevd_us <- fevd(var_us, n.ahead = 120)
cat("\nUS Cholesky FEVD at 120 months:\n")
print(fevd_us$US_Expectations[120,])
print(fevd_us$US_Bond[120,])

fevd_uk <- fevd(var_uk, n.ahead = 120)
cat("\nUK Cholesky FEVD at 120 months:\n")
print(fevd_uk$UK_Expectations[120,])
print(fevd_uk$UK_Bond[120,])

par(mfrow = c(1, 2))
plot(irf_us_exp, main = "US: Expectations Response", ylab = "Change in Expectations")
plot(irf_uk_exp, main = "UK: Expectations Response", ylab = "Change in Expectations")

par(mfrow = c(1, 2))
plot(irf_us_bond, main = "US: Bond Yield Response", ylab = "Change in Yield")
plot(irf_uk_bond, main = "UK: Bond Yield Response", ylab = "Change in Yield")
par(mfrow = c(1, 1))

## COVID subsample
cat("\n--- COVID subsample ---\n")
us_covid_ts <- tail(us_ts, 62)
uk_covid_ts <- tail(uk_ts, 62)

var_us_covid <- VAR(us_covid_ts, p = 2, type = "const")
var_uk_covid <- VAR(uk_covid_ts, p = 2, type = "const")

irf_us_covid_exp <- irf(var_us_covid, impulse = "US_Inflation", response = "US_Expectations",
                      n.ahead = 24, boot = TRUE, ci = 0.95)
irf_uk_covid_exp <- irf(var_uk_covid, impulse = "UK_Inflation", response = "UK_Expectations",
                      n.ahead = 24, boot = TRUE, ci = 0.95)

par(mfrow = c(1, 2))
plot(irf_us_covid_exp, main = "US: COVID Era Expectations", ylab = "Expectations Response")
plot(irf_uk_covid_exp, main = "UK: COVID Era Expectations", ylab = "Expectations Response")
par(mfrow = c(1, 1))

fevd_us_covid <- fevd(var_us_covid, n.ahead = 24)
fevd_uk_covid <- fevd(var_uk_covid, n.ahead = 24)
cat("\nUS COVID FEVD (24m):\n"); print(fevd_us_covid$US_Expectations[24,])
cat("\nUK COVID FEVD (24m):\n"); print(fevd_uk_covid$UK_Expectations[24,])

# sign restrictions on COVID subsample
us_covid_draws <- get_svar_draws(var_us_covid, n_draws = 500)
uk_covid_draws <- get_svar_draws(var_uk_covid, n_draws = 500)
B_us_covid_mtr <- select_mtr_draw(us_covid_draws, var_us_covid, n.ahead = 24)
B_uk_covid_mtr <- select_mtr_draw(uk_covid_draws, var_uk_covid, n.ahead = 24)

us_covid_fevd_mtr <- get_fevd_from_B(var_us_covid, B_us_covid_mtr, n.ahead = 24)
uk_covid_fevd_mtr <- get_fevd_from_B(var_uk_covid, B_uk_covid_mtr, n.ahead = 24)

cat("COVID US supply shock (24m):", round(us_covid_fevd_mtr[24,1]*100, 2), "%\n")
cat("COVID UK supply shock (24m):", round(uk_covid_fevd_mtr[24,1]*100, 2), "%\n")

## Pre-COVID baseline ----
cat("\n--- Pre-COVID baseline ---\n")
n_total <- nrow(us_ts)
us_precovid_ts <- head(us_ts, n_total - 62)
uk_precovid_ts <- head(uk_ts, n_total - 62)

```

```

p_us_pre <- as.numeric(VARselect(us_preCovid_ts, lag.max = 12, type = "const")$selection["SC(n)"])
p_uk_pre <- as.numeric(VARselect(uk_preCovid_ts, lag.max = 12, type = "const")$selection["SC(n)"])

var_us_preCovid <- VAR(us_preCovid_ts, p = p_us_pre, type = "const")
var_uk_preCovid <- VAR(uk_preCovid_ts, p = p_uk_pre, type = "const")

irf_us_pre_exp <- irf(var_us_preCovid, impulse = "US_Inflation", response = "US_Expectations",
  n.ahead = 120, boot = TRUE, ci = 0.95)
irf_uk_pre_exp <- irf(var_uk_preCovid, impulse = "UK_Inflation", response = "UK_Expectations",
  n.ahead = 120, boot = TRUE, ci = 0.95)

par(mfrow = c(1, 2))
plot(irf_us_pre_exp, main = "US: Pre-COVID Expectations", ylab = "Response")
plot(irf_uk_pre_exp, main = "UK: Pre-COVID Expectations", ylab = "Response")
par(mfrow = c(1, 1))

fevd_us_preCovid <- fevd(var_us_preCovid, n.ahead = 120)
fevd_uk_preCovid <- fevd(var_uk_preCovid, n.ahead = 120)
cat("\nUS pre-COVID FEVD:\n"); print(fevd_us_preCovid$US_Expectations[120,])
cat("UK pre-COVID FEVD:\n"); print(fevd_uk_preCovid$UK_Expectations[120,])

# sign restrictions on pre-COVID
us_pre_draws <- get_svar_draws(var_us_preCovid, n.draws = 500)
uk_pre_draws <- get_svar_draws(var_uk_preCovid, n.draws = 500)
B_us_pre_mtr <- select_mtr_draw(us_pre_draws, var_us_preCovid, n.ahead = 120)
B_uk_pre_mtr <- select_mtr_draw(uk_pre_draws, var_uk_preCovid, n.ahead = 120)

us_pre_fevd_mtr <- get_fevd_from_B(var_us_preCovid, B_us_pre_mtr, n.ahead = 120)
uk_pre_fevd_mtr <- get_fevd_from_B(var_uk_preCovid, B_uk_pre_mtr, n.ahead = 120)

cat("Pre-COVID US supply shock (120m):", round(us_pre_fevd_mtr[120,1]*100, 2), "%\n")
cat("Pre-COVID UK supply shock (120m):", round(uk_pre_fevd_mtr[120,1]*100, 2), "%\n")

## Summary of key numbers

cat("\n\n===== KEY RESULTS =====\n")
cat("Full sample FEVD (supply shock, 120m):\n")
cat("  US:", round(us_fevd_mtr[120,1]*100, 2), "%\n")
cat("  UK:", round(uk_fevd_mtr[120,1]*100, 2), "%\n")
cat("Pre-COVID (supply shock, 120m):\n")
cat("  US:", round(us_pre_fevd_mtr[120,1]*100, 2), "%\n")
cat("  UK:", round(uk_pre_fevd_mtr[120,1]*100, 2), "%\n")
cat("COVID era (supply shock, 24m):\n")

cat("  US:", round(us_covid_fevd_mtr[24,1]*100, 2), "%\n")
cat("  UK:", round(uk_covid_fevd_mtr[24,1]*100, 2), "%\n")
cat("Bond yield (Cholesky, 120m):\n")
cat("  US:", round(fevd_us$US_Bond[120,"US_Inflation"]*100, 2), "%\n")
cat("  UK:", round(fevd_uk$UK_Bond[120,"UK_Inflation"]*100, 2), "%\n")
cat("=====\n")

## Appendix tables

# Full FEVD across all shocks and variables (sign-restricted)
get_full_fevd <- function(var_model, B_draw, n.ahead = 120) {
  A_list <- Acoef(var_model)
  p <- var_model$p
  k <- ncol(B_draw)

  A_comp <- matrix(0, k*p, k*p)
  for (i in 1:p) A_comp[1:k, ((i-1)*k+1):(i*k)] <- A_list[[i]]
  if (p > 1) A_comp[(k+1):(k*p), 1:(k*(p-1))] <- diag(k*(p-1))

  MA <- array(0, dim = c(k, k, n.ahead+1))
  MA[,1] <- diag(k)
  companion_power <- diag(k*p)
  for (h in 1:n.ahead) {
    companion_power <- companion_power %*% A_comp
    MA[,h+1] <- companion_power[1:k, 1:k]
  }

  shock_var <- matrix(0, k, k)
  for (h in 1:(n.ahead+1)) {
    impulse_h <- MA[,h] %*% B_draw
    for (j in 1:k) {
      shock_var[,j] <- shock_var[,j] + impulse_h[j]^2
    }
  }
  total_var <- rowSums(shock_var)
  fevd_pct <- sweep(shock_var, 1, total_var, "/" ) * 100
  return(fevd_pct)
}

vnames <- c("Inflation", "Expectations", "Policy Rate", "Bond Yield")
snames <- c("Supply", "Demand", "Monetary Policy", "Residual")

```



```

# Table A1: full sample
cat("\n--- Table A1: Full sample FEVD (sign-restricted, MTR) ---\n")
cat("\nUS (120m):\n")
us_full_fevd <- get_full_fevd(var_us, B_us_mtr, 120)
rownames(us_full_fevd) <- vnames; colnames(us_full_fevd) <- snames
print(round(us_full_fevd, 2))

cat("\nUK (120m):\n")
uk_full_fevd <- get_full_fevd(var_uk, B_uk_mtr, 120)
rownames(uk_full_fevd) <- vnames; colnames(uk_full_fevd) <- snames
print(round(uk_full_fevd, 2))

# Table A2: pre-COVID
cat("\n--- Table A2: Pre-COVID FEVD (sign-restricted, MTR) ---\n")
cat("\nUS (120m):\n")
us_pre_full <- get_full_fevd(var_us_precovid, B_us_pre_mtr, 120)
rownames(us_pre_full) <- vnames; colnames(us_pre_full) <- snames
print(round(us_pre_full, 2))

cat("\nUK (120m):\n")
uk_pre_full <- get_full_fevd(var_uk_precovid, B_uk_pre_mtr, 120)
rownames(uk_pre_full) <- vnames; colnames(uk_pre_full) <- snames
print(round(uk_pre_full, 2))

# Table A3: COVID era
cat("\n--- Table A3: COVID-era FEVD (sign-restricted, MTR) ---\n")
cat("\nUS (24m):\n")
us_covid_full <- get_full_fevd(var_us_covid, B_us_covid_mtr, 24)
rownames(us_covid_full) <- vnames; colnames(us_covid_full) <- snames
print(round(us_covid_full, 2))

cat("\nUK (24m):\n")
uk_covid_full <- get_full_fevd(var_uk_covid, B_uk_covid_mtr, 24)
rownames(uk_covid_full) <- vnames; colnames(uk_covid_full) <- snames
print(round(uk_covid_full, 2))

# Table A4: Cholesky vs sign-restriction comparison
cat("\n--- Table A4: Cholesky vs sign-restricted (supply shock -> expectations) ---\n")

chol_us_full <- fevd_us$US_Expectations[120, "US_Inflation"] * 100
chol_uk_full <- fevd_uk$UK_Expectations[120, "UK_Inflation"] * 100
chol_us_pre <- fevd_us_precovid$US_Expectations[120, "US_Inflation"] * 100
chol_uk_pre <- fevd_uk_precovid$UK_Expectations[120, "UK_Inflation"] * 100
chol_us_cov <- fevd_us_covid$US_Expectations[24, "US_Inflation"] * 100
chol_uk_cov <- fevd_uk_covid$UK_Expectations[24, "UK_Inflation"] * 100

sr_us_full <- us_fevd_mtr[120, 1] * 100
sr_uk_full <- uk_fevd_mtr[120, 1] * 100
sr_us_pre <- us_pre_fevd_mtr[120, 1] * 100
sr_uk_pre <- uk_pre_fevd_mtr[120, 1] * 100
sr_us_cov <- us_covid_fevd_mtr[24, 1] * 100
sr_uk_cov <- uk_covid_fevd_mtr[24, 1] * 100

comparison <- data.frame(
  Sample = c("Full (1996-2025)", "Full (1996-2025)",
    "Pre-COVID (1996-2019)", "Pre-COVID (1996-2019)",
    "COVID-Era (2020-2025)", "COVID-Era (2020-2025)"),
  Country = rep(c("US", "UK"), 3),
  Cholesky = round(c(chol_us_full, chol_uk_full, chol_us_pre, chol_uk_pre,
    chol_us_cov, chol_uk_cov), 2),
  Sign_Restricted = round(c(sr_us_full, sr_uk_full, sr_us_pre, sr_uk_pre,
    sr_us_cov, sr_uk_cov), 2)
)
print(comparison)

# Table A5: full Cholesky FEVD
cat("\n--- Table A5: Cholesky FEVD, full sample ---\n")
cat("\nUS (120m):\n")
print(round(fevd_us$US_Inflation[120,] * 100, 2))
print(round(fevd_us$US_Expectations[120,] * 100, 2))
print(round(fevd_us$US_Interest[120,] * 100, 2))
print(round(fevd_us$US_Bond[120,] * 100, 2))

cat("\nUK (120m):\n")
print(round(fevd_uk$UK_Inflation[120,] * 100, 2))
print(round(fevd_uk$UK_Expectations[120,] * 100, 2))
print(round(fevd_uk$UK_Interest[120,] * 100, 2))
print(round(fevd_uk$UK_Bond[120,] * 100, 2))

```

```

# Demand and monetary policy IRFs (not main focus but included for completeness)
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))

plot(0:120, sirf_us[2, 2, ], type = "l", col = "blue", lwd = 2,
     main = "US: Expectations to Demand Shock", xlab = "Months", ylab = "Response")
abline(h = 0, lty = 2, col = "grey50")

plot(0:120, sirf_uk[2, 2, ], type = "l", col = "red", lwd = 2,
     main = "UK: Expectations to Demand Shock", xlab = "Months", ylab = "Response")
abline(h = 0, lty = 2, col = "grey50")

plot(0:120, sirf_us[4, 3, ], type = "l", col = "blue", lwd = 2,
     main = "US: Bond Yield to MP Shock", xlab = "Months", ylab = "Response")
abline(h = 0, lty = 2, col = "grey50")

plot(0:120, sirf_uk[4, 3, ], type = "l", col = "red", lwd = 2,
     main = "UK: Bond Yield to MP Shock", xlab = "Months", ylab = "Response")
abline(h = 0, lty = 2, col = "grey50")

par(mfrow = c(1, 1))

# IRF credible sets across all 500 draws (68% and 90%)
compute_irf_range <- function(draws, var_model, resp_var, shock_col, n.ahead) {
  all_irfs <- matrix(NA, nrow = length(draws), ncol = n.ahead + 1)
  for (d in seq_along(draws)) {
    irf_d <- get_structural_irf(var_model, draws[[d]], n.ahead)
    all_irfs[d, ] <- irf_d[resp_var, shock_col, ]
  }
  list(
    median = apply(all_irfs, 2, median),
    lo16 = apply(all_irfs, 2, quantile, 0.16),
    hi84 = apply(all_irfs, 2, quantile, 0.84),
    lo05 = apply(all_irfs, 2, quantile, 0.05),
    hi95 = apply(all_irfs, 2, quantile, 0.95)
  )
}

cat("\nComputing IRF ranges across draws...\n")
us_range <- compute_irf_range(us_draws, var_us, 2, 1, 120)
uk_range <- compute_irf_range(uk_draws, var_uk, 2, 1, 120)

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
h <- 0:120

plot(h, us_range$median, type = "l", col = "black", lwd = 2,
     ylim = range(c(us_range$lo05, us_range$hi95)),
     main = "US: Supply Shock -> Expectations\n(500 draws)",
     xlab = "Months", ylab = "Response")
polygon(c(h, rev(h)), c(us_range$lo05, rev(us_range$hi95)),
        col = rgb(0.8, 0.2, 0.2, 0.15), border = NA)
polygon(c(h, rev(h)), c(us_range$lo16, rev(us_range$hi84)),
        col = rgb(0.8, 0.2, 0.2, 0.3), border = NA)
lines(h, us_range$median, col = "black", lwd = 2)
abline(h = 0, lty = 2, col = "grey50")
legend("topright", c("Median", "68%", "90%"),
       col = c("black", rgb(0.8, 0.2, 0.2, 0.3), rgb(0.8, 0.2, 0.2, 0.15)),
       lwd = c(2, 8, 8), cex = 0.7, bty = "n")

plot(h, uk_range$median, type = "l", col = "black", lwd = 2,
     ylim = range(c(uk_range$lo05, uk_range$hi95)),
     main = "UK: Supply Shock -> Expectations\n(500 draws)",
     xlab = "Months", ylab = "Response")
polygon(c(h, rev(h)), c(uk_range$lo05, rev(uk_range$hi95)),
        col = rgb(0.2, 0.2, 0.8, 0.15), border = NA)
polygon(c(h, rev(h)), c(uk_range$lo16, rev(uk_range$hi84)),
        col = rgb(0.2, 0.2, 0.8, 0.3), border = NA)
lines(h, uk_range$median, col = "black", lwd = 2)
abline(h = 0, lty = 2, col = "grey50")
legend("topright", c("Median", "68%", "90%"),
       col = c("black", rgb(0.2, 0.2, 0.8, 0.3), rgb(0.2, 0.2, 0.8, 0.15)),
       lwd = c(2, 8, 8), cex = 0.7, bty = "n")

par(mfrow = c(1, 1))

# FEVD distribution across draws (supply shock -> expectations)
get_fevd_expectations_supply <- function(var_model, B_draw, n.ahead) {
  A_list <- Acoef(var_model)
  p <- var_model$p; k <- ncol(B_draw)
  A_comp <- matrix(0, k*p, k*p)
  for (i in 1:p) A_comp[1:k, ((i-1)*k+1):(i*k)] <- A_list[[i]]
  if (p > 1) A_comp[(k+1):(k*p), 1:(k*(p-1))] <- diag(k*(p-1))

  shock_var_exp <- 0
  total_var_exp <- 0
  companion_power <- diag(k*p)
  MA_h <- diag(k)

```



```

impulse_0 <- MA_h %>% B_draw
shock_var_exp <- impulse_0[2, 1]^2
total_var_exp <- sum(impulse_0[2, ]^2)

for (h in 1:n.ahead) {
  companion_power <- companion_power %>% A_comp
  MA_h <- companion_power[1:k, 1:k]
  impulse_h <- MA_h %>% B_draw
  shock_var_exp <- shock_var_exp + impulse_h[2, 1]^2
  total_var_exp <- total_var_exp + sum(impulse_h[2, ]^2)
}
return(shock_var_exp / total_var_exp * 100)
}

cat("\n--- Table A6: FEVD uncertainty across 500 draws ---\n")

us_fevd_dist <- sapply(us_draws, function(B) get_fevd_expectations_supply(var_us, B, 120))
uk_fevd_dist <- sapply(uk_draws, function(B) get_fevd_expectations_supply(var_uk, B, 120))
cat("Full sample (120m):\n")
cat(sprintf("  US: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(us_fevd_dist), quantile(us_fevd_dist, 0.16), quantile(us_fevd_dist, 0.84)))
cat(sprintf("  UK: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(uk_fevd_dist), quantile(uk_fevd_dist, 0.16), quantile(uk_fevd_dist, 0.84)))

us_pre_fevd_dist <- sapply(us_pre_draws, function(B) get_fevd_expectations_supply(var_us_precovid, B, 120))
uk_pre_fevd_dist <- sapply(uk_pre_draws, function(B) get_fevd_expectations_supply(var_uk_precovid, B, 120))
cat("Pre-COVID (120m):\n")
cat(sprintf("  US: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(us_pre_fevd_dist), quantile(us_pre_fevd_dist, 0.16), quantile(us_pre_fevd_dist, 0.84)))
cat(sprintf("  UK: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(uk_pre_fevd_dist), quantile(uk_pre_fevd_dist, 0.16), quantile(uk_pre_fevd_dist, 0.84)))

us_cov_fevd_dist <- sapply(us_covid_draws, function(B) get_fevd_expectations_supply(var_us_covid, B, 24))
uk_cov_fevd_dist <- sapply(uk_covid_draws, function(B) get_fevd_expectations_supply(var_uk_covid, B, 24))
cat("COVID era (24m):\n")
cat(sprintf("  US: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(us_cov_fevd_dist), quantile(us_cov_fevd_dist, 0.16), quantile(us_cov_fevd_dist, 0.84)))
cat(sprintf("  UK: median %.2f%%, [16th: %.2f%%, 84th: %.2f%%]\n",
  median(uk_cov_fevd_dist), quantile(uk_cov_fevd_dist, 0.16), quantile(uk_cov_fevd_dist, 0.84)))

# Table A7: lag selection details
cat("\n--- Table A7: lag selection ---\n")
cat("\nUS full sample:\n"); print(lag_select_us$criteria)
cat("\nUK full sample:\n"); print(lag_select_uk$criteria)
cat("\nPre-COVID lags (BIC): US =", p_us_pre, ", UK =", p_uk_pre, "\n")
cat("COVID era: fixed at 2 lags (short sample)\n")

```